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The use of two-dimensional materials in high-temperature rechargeable batteries: Current issues and preventative measures

Running title: The use of 2D materials in Li-based batteries

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Abstract

As their name suggests, energy storage systems are used for the on-demand supply and storage of energy. However, the development of high-capacity energy storage systems that can safely operate at high temperatures remains a challenge, as exemplified by the occurrence of thermal runaway in rechargeable batteries, especially in harsh environments. In view of the above, the increasing demand for rechargeable (e.g., Li-ion) batteries necessitates the development of safer power sources operable at high temperatures and requires the collective contribution of chemists, materials scientists, and engineers to gain a deep understanding of the thermal runaway mechanism and take preventative measures. Currently, thermal runaway is prevented by the use of safety devices and/or inherent safety methods. The latter approach involves the design of electrodes and electrolytes (including the separator) to increase the thermal stability of rechargeable batteries and is faced by the challenge of finding a compromise between electrochemical performance and thermal stability. The discovery of graphene and its outstanding properties gave rise to a new class of two-dimensional (2D) materials that are believed to offer a solution to the above problem and have therefore attracted increased research attention. However, the applications of most non-graphene 2D materials have been underexplored, since these materials have been discovered only very recently, and the corresponding research has mainly focused on investigating their properties and synthesis routes. In this review, we outline the progress in the development of safe and high-performance rechargeable batteries based on 2D materials, focusing on sustainable battery operation under high-temperature conditions.

Keywords: two-dimensional, nanocomposite, Li-ion battery, Li-S battery, thermal runaway

1. Introduction

Li-ion batteries (LIBs) were first commercialized by Sony Corporation in 1991 and currently dominate the market of rechargeable batteries. Featuring the reversible (de)intercalation of Li ions and exhibiting the combined advantages of relative safety (offered by the utilization of a graphite anode hosting Li in the form of Li_xC_6), high voltage (3.7 V), and the use of high-energy cathode material ($\text{Li}_{1-x}\text{CoO}_2$) and a non-aqueous electrolyte.^{1,2} Although the above properties make LIBs popular multi-purpose energy storage units having a large market share in the areas of electric vehicles and electronic devices,³ the use of a graphite anode with a low theoretical specific capacity of only 372 mAh/g lowers the energy density of LIBs. Much effort has been directed at increasing the energy density of LIBs to meet the ever-increasing demands of the energy storage market. To date, front-end LIBs exhibit high theoretical gravimetric and volumetric energy densities of up to 260 Wh/kg and 770 Wh/l, respectively.⁴⁻⁶ Compared to that of other metals, the utilization of Li as an anode material is very advantageous in view of its high theoretical capacity of 3861 mAh/g,⁷ low molecular weight (6.94 g/mol), and the most negative reduction potential among all metals (−3.04 V vs. the standard hydrogen electrode).⁸ Although the superior energy densities of LIBs have revolutionized energy storage, these batteries still need to be improved in terms of cost, cycle lifetime, safety, and operating temperature range. Moreover, a number of application fields, including oil, defense, automotive, and aerospace industries place very strict requirements on the robustness of rechargeable energy storage systems, which commonly suffer from the disadvantage of high materials cost and can therefore not be widely utilized in such complex applications.⁹ The high abundance of Li in the Earth's crust implies that we are unlikely to experience a Li shortage and suggests that the high cost of LIBs can be mainly ascribed to the use of expensive transition metals (e.g., Co) in cell components.¹⁰ However, the safety and the limited operating temperature range of LIBs are still a major area of concern. Specifically, the highest risk is posed by the use of the Li anode and arises from the temperature increase – induced growth of Li dendrites after repeated charge-discharge cycling that can lead to thermal runaway. To date, this issue has mostly been addressed by preventing Li dendrite growth and thus increasing the application scope of LIBs to a wider range of operating conditions.¹¹ The performance, lifetime, and safety

of LIBs are highly dependent on operating temperature, and operation at temperatures outside the normal range can cause catastrophic failures leading to thermal runaway events.¹²

In 2020, the global high-temperature energy storage market is projected to be valued at \$2.5 billion, whereas the oil-drilling battery market was valued at \$100 million in 2011.¹³ Importantly, downhole power sensors require the use of the corresponding energy storage systems such as primary batteries, secondary batteries, fuel cells, and capacitors, whereas energy storage systems used in the oil industry must withstand a variety of operating temperatures (up to 150 °C and higher).¹⁴ High-temperature electronics are also used in the aviation industry, especially in systems operating in close proximity to the engine, e.g., close-to-actuator power electronics in motor controls and brake systems face temperatures around 350 °C, while distributed control systems close to the engine face temperatures of –55 to 250 °C. The aerospace industry also needs high-temperature energy storage systems to power devices exploring extraterrestrial surfaces,¹³ and the high-temperature battery market are thought to feature a significant contribution from the hybrid vehicle application field, since engine, transmission, and brake systems require electronics to operate under elevated temperatures ranging from 200 to 800 °C. In addition, electric vehicles rely on energy sourced from batteries rather than from fossil fuels, thereby requiring the corresponding systems to operate at high temperatures.¹³

Currently, Li thionyl chloride (Li-SOCl₂) primary batteries are commonly used as downhole energy storage systems. Compared to other primary batteries, the above systems offer high operating temperatures (up to 125 °C), high operating potentials (up to 3.6 V), excellent volumetric energy densities (up to 1280 Wh/l), and increased power density, additionally featuring the advantages of long operational lifetime and low self-discharge rate. However, Li-SOCl₂ batteries also feature a number of disadvantages, e.g., they are not rechargeable and can explode when damaged, which raises environment, health, and safety concerns.¹⁴ To overcome these drawbacks, FastCAP Systems introduced rechargeable, high-temperature (up to 150 °C), and safe ultracapacitors with high power densities as an alternative to batteries,¹⁵ subsequently developing an ultracapacitor suited for operation at temperatures of ≥250 °C.¹⁶ Moreover, researchers working on a joint project between Evolution Engineering (Canada) and two German institutes (Custom Cells Itzehoe GmbH & Fraunhofer Institute for Silicon Technology) are currently developing a

rechargeable energy storage system for downhole applications that is expected to perform safely at temperatures of 150 °C or higher.¹⁷ Caja *et al.* reported the development of a CC-size rechargeable Li-ion molten salt battery that can be operated at temperatures of up to 150 °C and exhibits a low voltage of 2.3 V and a capacity of 2.2 Ah. The above battery was reported to exhibit coulombic efficiencies of 99 and 80% at 100 and 120 °C, respectively, with further efficiency and lifetime decreases observed with increasing temperature.¹⁸ While research on high-temperature rechargeable batteries is still in progress, some of these batteries are already available on the market. Specifically, Eagle Picher Technology LLC reported the development of safe D and sub-DD lithium carbon monofluoride (Li/CF_x) batteries that are suitable for oil industry applications and show high thermal stability (up to 160 °C), good discharge efficiency, and high rate capability.^{19,20} Moreover, Electrochemical Systems, Inc. developed downhole batteries for operation at temperatures above 130 °C that contain flame-resistant and non-explosive molten salt electrolytes and thus exhibit an improved safety profile.²¹ Figure 1 summarizes the classification of high-temperature batteries available on the market and under development.

Recently, the use of two-dimensional (2D) materials has enabled the fabrication of high-temperature batteries and allowed the maximization of rechargeable battery performance with minimal compromise on safety and stability. In view of their enhanced surface interaction and material properties, 2D materials have also shown potential for use in batteries operating in extreme environments. As exemplified by the recently developed (Rice University) rechargeable batteries (featuring an electrolyte/separator made up of hexagonal boron nitride, an ionic liquid, and a lithium salt) that can operate at temperatures of up to 150 °C and exhibit enhanced electrochemical stability and an acceptable working potential. Notably, all electrolyte components in the above batteries are non-flammable and safe to use.^{22,23}

This review focuses on the properties, mechanism, and progress of the integration of 2D materials into high-temperature rechargeable batteries, additionally discussing the demands, challenges and current development of these batteries for operation in extreme environments.

2. 2D materials in LIBs

Since the performance of rechargeable batteries is influenced by battery cell components, e.g., the electrodes and the electrolyte,²⁴ much effort is directed at the development of new cell components to push the boundaries of energy density, cycle lifetime, cost, and safety as well as to widen the operating temperature range. In view of the fact that material properties are greatly influenced by size and dimensionality, low-dimensional materials commonly exhibit unique properties (due to the confinement effect) that are not observed for their bulk counterparts. Recently, such materials have been employed in the electrodes and other components of LIBs to enhance their performance.⁴ The use of nanostructures as electrodes has enabled the introduction of new Li storage mechanisms,^{25,26} allowing one to decrease the path lengths²⁷ of Li ion and electron transport and thus increase charge/discharge rates.^{28,29} Moreover, the above electrodes exhibited stable cycling performance, since the utilization of nanomaterials decreased volume expansion/contraction and preserved electrode integrity.^{26,30–32} Another advantage of employing nanomaterials is the flexibility of modulating the working potential,^{29,33} and materials electrochemically inactive in bulk were found to be highly efficient in storing Li at the nanoscale because of the presence of partially bonded atoms in nanomaterials and their high surface energies.²⁹ In particular, nanomaterials can be designed in different shapes (such as rods, particles, cubes, and plates) and dimensions, which offers high and easily available Li ion storage sites and results in enhanced electrochemical performance.²⁸

Dimensionality is a key requirement for designing materials at the nanoscale, as it not only determines the atomic structure of a given material but also defines its properties.³⁴ According to dimensionality, materials can be divided into zero, one-, two-, and three-dimensional ones (0D, 1D, 2D, and 3D, respectively). Specifically, 0D materials are those with all three dimensions confined to the nanoscale, whereas low-dimensional materials are systems in which the electronic-state wavefunction is confined in at least one of the three dimensions, i.e., where one of the dimensions is in the range of 1–100 nm. On the other hand, bulk materials are classified as 3D materials with no confinement effects. Dimensionality alters the electronic properties of materials. In 0D materials (quantum dots), electrons are confined in three directions, i.e., cannot easily move in any dimension and therefore behave as atomic/molecular clusters with discrete states that are well separated in energy. In 1D materials (quantum wires), electrons are confined in two directions, easily moving in one dimension. In 2D materials (thin films), electrons are confined in one

direction and therefore exist in quantum wells, easily moving in two dimensions. With increasing excitation energy, the density of states of 2D materials exhibits a quasi-continuous step-like increase.^{34,35}

Nanomaterials with different dimensionalities (particularly 0D, 1D, and 3D materials) have recently been employed as battery electrodes. However, 0D, 1D, and 3D nanomaterials feature poor charge behavior unless they have a porous structure or are part of composites. The main drawback of the corresponding electrodes is that the whole electrode cannot be involved in the electrochemical reaction, as ions cannot diffuse throughout the electrode body, which results in low electrochemical performance. In addition, the capacity of charge storage devices is strongly affected by the electrode surface area, which is limited in 0D, 1D, and 3D materials. To overcome these limitations, the use of 2D materials in electrodes was extensively investigated in the past decade, and the above materials were found to exhibit the advantages of high specific surface area³⁶ and whole-electrode-body accessibility. The high surface-to-volume ratios, short diffusion paths, high mechanical strengths, and good conductivities of 2D materials account for their high specific Li storage capacity and allow them to accommodate more Li ions during intercalation, making these materials and their composites excellent candidates for use in charge storage devices such as Li-based rechargeable batteries³⁷. However, research on 2D materials is still in its infancy, e.g., although the use of 2D materials in rechargeable batteries has been investigated, most research efforts have focused on understanding the unique properties of these materials.

Notably, quantum confinement³⁸ and high surface area endow 2D materials with exceptional mechanical, chemical, and optical properties, thereby enabling their use in a wide range of applications such as energy storage devices.^{39–41} 2D materials can be classified as either 2D allotropes or 2D compounds.³⁸ The former comprise 2D structures containing a single element, e.g., C (graphene), Si (silicene), Ge (germanene), and P (phosphorene),^{42–45} whereas the latter feature covalent networks comprising two elements, as exemplified by hexagonal boron nitride (h-BN), silicon carbide (SiC), transition metal dichalcogenides (MoS₂, MoSe₂, WS₂, WSe₂, etc.), transition metal oxides (LaVO₃, LaMnO₃), transition metal chalcogenides (NbSe₃, TaSe₃), MXenes (transition metal carbides, nitrides, and carbonitrides), layered complex oxides, and others (Li₇MnP₄, MnP₄).

With graphene as a representative of the 2D allotropes, graphene's exceptional and unusual properties compared to those of its bulk counterpart (graphite), attracting increased attention since 2004.^{46–49} In terms of electronic properties, graphene is a zero-bandgap semiconductor with high carrier mobility.^{50–54} The thermal conductivity of graphene measured at room temperature ranges from 2600 to 5300 W/mK and is therefore extremely high, exceeding those of all known materials.⁵⁵ Graphene possesses excellent mechanical properties, featuring a breaking strength and a Young's modulus of 42 Nm and 1.0 TPa (cf. values of single-layer MoS₂ nanosheets (~270 GPa), bulk MoS₂ (~240 GPa), and steel (~205 GPa)), respectively, and therefore being the strongest material known to date. Moreover, graphene can sustain elastic deformations exceeding 20% and features outstanding electronic properties typical of covalently bound 2D materials.⁴⁸ Considering these physical properties, the use of graphene has risen rapidly in energy storage applications in the recent years.

The extraordinary properties of graphene inspired research on other 2D materials. Hexagonal boron nitride (h-BN)^{56–58} with an analogous structure to graphite also known as white graphite^{59–62} is a wide-bandgap semiconductor (5–6 eV), making it a good candidate as electrolyte and separator parts of the battery where electrical conduction is not required. Compared to carbon materials, h-BN or BN structures generally have higher thermal conductivities (k , i.e., the flow of heat per unit area to the temperature gradient).^{59–61,63,64} Specifically, h-BN nanosheets were shown to exhibit k values of 300–2000 Wm/K,⁶¹ which are comparable to those of graphene (2000–4000 Wm/K).^{61,65–67}

More recently, transition metal dichalcogenides (TMDs, mainly sulfides, and selenides) have attracted attention as energy storage devices due to their tunable band gap, unique layered structure, and rich redox sites making it convenient for the intercalation and transportation of Li⁺ ions.^{68,69} TMDs have variable band gaps, insulators such as HfS₂, semiconductors such as SnS₂ and MoS₂, semi-metals such as TiS₂ and WTe₂, and true metals such as TaS₂ and VSe₂.^{70–72} The electronic structure of TMDs are tunable and unlike graphene, the well-defined structure of TMDs is not affected by doping. These tunable properties make them ideal materials for electronics and

electrochemical devices. Layered TMDs also possess catalytic properties, a property essential for electrochemical devices that require accelerated kinetics.^{73,74}

Moreover, as a large family of new type 2D transition metal carbides/carbonitrides materials (MXenes), have been widely developed in energy storage since the first discovery of Ti_3C_2 in 2011. The first 2D layered MXene, was isolated in 2011.⁷⁵ MXenes are prepared by selective etching of certain atomic layers from their layered precursors, as known as MAX phase with a formula of $\text{M}_{n+1}\text{AX}_n$ ($n = 1, 2, 3$). Where M represents early transition metals ($\text{M} = \text{Ti, Sr, V, Cr, Ta, Nb, Zr, Mo, or Hf}$), A represents main-group sp elements (mostly IIIA or IVA), and X represents either C or N as well as both.⁷⁶ MXenes possess excellent conductivity, with 2D structure and Lewis acid surface that not only store large amounts of electrical charges but also discharge at rapid rates with reversible expansion and contraction during charge/discharge. This makes MXenes great potentials in energy storage applications.^{77,78}

The outstanding mechanical, electrical, and thermal properties of 2D materials make them increasingly attractive battery components capable of expanding the range of operating conditions, especially for high-temperature applications. This trend has been highlighted in the upcoming sections, which also present a compilation of studies on the use of these materials in rechargeable batteries.

3. 2D material-based composites for high-temperature LIBs: Issues and preventative measures

The use of 2D materials in batteries has been shown to enhance their performance and stability significantly. This section attempts to review studies performed to date on the effects of 2D on the safety, performance, and stability of LIBs under high-temperature conditions.

Thermal runaway is a major safety issue of LIBs. Conditions such as elevated temperature, high charge levels, or overcharging result in the abuse of one or more battery components, causing short-circuiting and thus leading to excess heat generation, fire, or explosion, which is referred to as thermal runaway and mainly occurs at electrodes and in electrolytes. A number of thermal runaway prevention measures have been investigated, including the use of safety devices and

inherently safe battery components (particularly electrodes, electrolytes, and the separator). Specifically, safety devices are designed to release the high pressure and heat prior to thermal runaway occurrence and thus present an engineering challenge. However, the inherent safety of battery components such as electrodes, electrolytes, and separators is a matter of even greater concern. Table 1 presents a brief overview of the safety issues associated with different LIB components, allowing us to consider detailed thermal runaway mechanisms in LIB components in addition to the measures taken to increase the inherent safety of batteries with emphasis on the use of 2D materials.

3.1.Cathode: Current issues and preventative measures using 2D materials

The most commonly used LIB cathode materials are layered LiMO_2 , Li-rich layered oxides ($\text{Li}_{1+x}\text{M}_{1-x}\text{O}_2$), spinels (LiM_2O_4), olivines (LiMPO_4), silicates (Li_2MSiO_4), borates (LiMBO_3) ($\text{M} = \text{Fe}, \text{Co}, \text{Ni}, \text{Mn}, \text{Cu}, \text{Cr}, \text{etc.}$), LiV_3O_8 , metal fluorides (FeF_3), and metal oxides (V_2O_5 , MnO_2).^{79,80} Thermal runaway at the cathode is represented by two degradation mechanisms,⁸¹ namely (i) cathode decomposition at high temperatures, which is accompanied by the evolution of gaseous byproducts and generates excess pressure inside the battery and (ii) reactions of the cathode with the electrolyte solvent at high temperatures.⁸²

Considering safety issues, coating of the cathode with materials that do not react with the electrolyte or degrade at high temperatures is the best way of increasing thermal stability. In LIBs, deterioration of the cathode surface induced by its reactions with the electrolyte can result in the formation of highly resistive passivation films. The application of a coating layer can inhibit this structural disorder by preventing the corresponding phase transition, thereby improving structural stability. In addition, the use of a coating layer prevents the direct contact of the cathode with the electrolyte and thus helps to avoid the undesired reactions and oxidation of the latter.

Thermal runaway at the cathode has been rather extensively investigated, with most studies focusing on the use of inorganic cathode coatings such as metal oxides (e.g., Al_2O_3 and ZnO), metal phosphates (e.g., AlPO_4), and metal fluorides (e.g., AlF_3) in view of their high stability. However, as these insulating materials increase the internal resistivity of LIBs, a compromise between safety and performance needs to be found. Sun *et al.* studied the effects of coating a Li-

enriched Ni-Mn oxide ($\text{Li}[\text{Li}_{0.19}\text{Ni}_{0.16}\text{Co}_{0.08}\text{Mn}_{0.57}]\text{O}_2$) cathode by AlF_3 , showing that although coating significantly improved the thermal stability of the cathode, it did not markedly improve its electrochemical performance, which was particularly obvious at higher contents of AlF_3 because of the insulating nature of this fluoride.⁸³

Coating of LIB cathodes with carbon materials (graphite, carbon nanotubes, etc.) has recently been introduced as a way to address the drawbacks of using metal oxide-based coatings. Specifically, the use of carbon not only prevents cathode surface deterioration but also increases cathode conductivity. For example, Mun *et al.* coated a $\text{Li}_{1.17}\text{Ni}_{0.17}\text{Co}_{0.11}\text{Mn}_{0.56}\text{O}_2$ cathode by a 100-nm-thick layer of multi-walled carbon nanotubes by applying shear stress without destroying the crystal structure of the electrode.⁸⁴

Although the current modifications make it possible to achieve safe and higher energy density cathode materials still, the cycling stability of the LiFePO_4 remains a challenge specifically when cycled at high current rates.^{85,86}

Moreover, carbon coating is difficult for materials such as $\text{LiNi}_{0.5}\text{Mn}_{1.5}\text{O}_4$, because a carbon source is needed to introduce carbon and thus its prepared under reducing atmosphere. However, $\text{LiNi}_{0.5}\text{Mn}_{1.5}\text{O}_4$ needs an oxygen atmosphere to avoid too many oxygen vacancies as it can negatively affect cycling stability and overall performance of the cathode.⁸⁷ Thus, it is important to pursue new cathode modification methods. To address this issue, 2D materials based on graphene and its derivatives have been proposed as alternative coating layers for cathodes in LIBs⁸⁸.

Nanohybrids of graphene-decorated carbon-coated LiFePO_4 (LFP@C/G) composite have been reported by Wang *et al.* as cathode materials for LIBs. The LFP@C/G composite was made via ball-milling combined with solid-state reaction. The as-prepared LFP@C/G composite with a 3 wt% graphene content resulted in a unique 3D "sheets-in-pellets" and "pellets-on-sheets" conducting network structure with abundant of mesopores. This structure would facilitate electronic and ionic transport. The LFP@C/G composite exhibited a high reversible capacity of 103.31 mAh/g after 500 cycles at 10 C.⁸⁹

Graphene has also been used to coat nickel-rich layered cathode materials (NCMs). Son *et al.* synthesized a graphene ball (GB) which is made of graphene-silica assembly prepared through chemical vapor deposition process. Results showed that only 1 wt% of GB was able to uniformly coat NCMs which was obtained through a mild Nobilta milling process. The GB coated NCMs as cathodes displayed reversible capacities of 205.78 mAh/g and 195.71 mAh/g, at 25 °C and 60 °C, respectively after 100 cycles at 0.1 C and a voltage cut-off of 4.5 V which was higher when compared to pristine NCMs. This could be attributed to the hierarchical three-dimensional structure of GB coating.⁹⁰

Beyond graphene, graphene oxide has also been applied as a coating for cathodes in LIBs. For example, Ma, Yuxin, *et al.* used 2.5 wt% of graphene oxide (GO) to coat $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ (NCM) through a simple approach which involved mixing of GO solution with NCM. The GO coating layer could act as a protection layer and thus suppressing unwanted reactions of the active material with the electrolyte. As a result, The GO-coated NCM cathodes displayed a high reversible capacity of 136.8 mAh/g after 100 cycles at 1 C.⁹¹

Reduced graphene oxide has also been used to wrap and/or coat $\text{LiNi}_{0.6}\text{Co}_{0.2}\text{Mn}_{0.2}\text{O}_{0.2}$ (NCM) cathodes as in the study reported by Shim, Jae-Hyun, *et al.*. The SEM images (Figure 2 (a-d)) and HRTEM images (Figure 2 (e-f)) clearly show that the NCM is completely wrapped by very thin and transparent rGO sheets as in shown in the schematic illustration (Figure 2 (g)). These rGO-wrapped NCMs as cathodes displayed a high reversible capacity of 179.9 mAh/g after 100 cycles at a current of 1 C which is higher than pristine NCMs (Figure 2 (h)). The high electrochemical performance is ascribed to the effectively suppressed detrimental side reactions at the cathode electrode.⁹²

Besides cathode coating, graphene has also been used as composites with cathode materials in LIBs. In a recent study by Fu, Yanqing, *et al.*, graphene formed into a composite with lithium iron phosphate (LFP) cathode (LFP/G). Results showed that the addition of 1 % of wrinkled and/or large graphene sheets could act as an effective conductive matrix without affecting the olivine

structure of LFP. The LFP/G electrodes exhibited a reversible capacity of 142 mAh/g after 150 cycles at a current rate of 1 C.⁹³

Other 2D materials beyond graphene and its derivatives which including h-BN, TMDs, MXenes, etc. have not been explored as coating layers or into composites with cathodes in LIBs due to the low voltage window versus Li/Li⁺. However, the intrinsic features of 2D materials beyond graphene makes them worth investigation as cathodes in LIBs.^{77,94}

In consideration of that, Wang, Chueh-Han, *et al.*, recently introduced MXenes as current collectors for cathodes. Due to the highest electrical conductivity of Ti₃C₂T_x among the MXenes family, Ti₃C₂T_x was used as a current collector (d-Ti₃C₂T_x film) in LIBs. Good compatibility and connection between d-Ti₃C₂T_x film and LiFePO₄ was achieved. Consequently, the d-Ti₃C₂T_x film based LiFePO₄ was able to regain 40 % of its initial capacity when measured in the range of 0.15 C–6 C rates displaying a high reversible capacity of 155 mAh/g after 70 cycles when returned to the initial current rate of 0.15 C. In view of these outstanding performances, the current collector was also used for anodes in LIBs.⁹⁵

3.2. Anode: Current issues and preventative measures using 2D materials

Although Li metal, exhibiting the highest possible theoretical specific capacity, is the first-choice material for LIB anodes, the formation of Li dendrites can cause thermal runaway and thus presents a number of challenges to be addressed.⁹⁶ The degradation mechanisms of the anode are similar to those of the cathode and feature the thermal degradation of the anode material and the occurrence of electrode-electrolyte reactions leading to thermal runaway. LIB charging at high voltages can initiate Li⁺ plating on the anode accompanied by dendrite formation, which results in irreversible capacity loss followed by a decrease in battery performance and thermal runaway. On the other hand, a solid-electrolyte interphase layer (SEI) comprising Li oxide, Li fluoride, and Li semicarbonates (e.g., lithium alkyl carbonates) can be formed at the anode. As the temperature is increased, alkyl carbonates found in the electrolyte decompose to form insoluble Li₂CO₃, which increases the thickness of the SEI layer and limits the efficiency of the anode, as it increases cell resistance and reduces capacity. Electrolyte decomposition also results in the evolution of gaseous products, increasing the internal pressure and leading to thermal runaway.

A number of techniques for mitigating the above safety issues have been established, including electrolyte modification with fluorinated compounds to stabilize Li anodes, the use of self-healing electrostatic shields, high salt concentrations, and the synergistic effect of Li polysulfides and LiNO_3 . Moreover, the utilization of solid electrolytes has been demonstrated to offer increased protection and prevent dendrite propagation.^{96,97}

However, the above-mentioned strategies do not prevent the large volume variation of the Li metal layer during cycling, therefore allowing lithium dendrite growth. In this regard, the anode volume change can be minimized by employing stable host materials, which results in guided Li plating. However, the conventionally used host materials are non-conductive hosts and are based on polymer gels or glass fibers and easily tend to lose electric contact.⁹⁸

To this extent, a mechanically stable material to account for the volume expansion and at the same time chemically stable to prevent any unwanted reactions towards lithium are the requirements for host materials. 2D materials are promising candidates, owing to their high mechanical strength, chemical stability, and flexibility.

The Li anode can be stabilized by the use of artificial SEIs made up of 2D materials. 2D materials including graphene and h-BN are believed to be particularly well suited for this purpose in view of their high stability toward aggressive chemicals such as Li. The mechanical strength of both graphene and h-BN significantly exceeds that of Li, which allows these materials to sustain the overpressure inside the battery cell resulting from Li dendrite formation. The diameter of graphene and h-BN pores is smaller than that of Li ions and other electrolyte components, which implies that the above 2D materials do not allow any of these components to pass through. Moreover, the high flexibility of mono- and few-layer graphene and h-BN allows them to accommodate Li deposition.

Yan *et al.* for the first time employed 2D graphene and h-BN to prevent Li dendrite formation, showing that the use of either graphene or h-BN in a Li metal anode leads to safer and more stable cycling performance, additionally revealing the occurrence of SEI alteration. The use of an h-BN–

modified Li metal anode resulted in a coulombic efficiency of 97% over 50 cycles and afforded cell voltages of up to 4.2 V even at a high operating temperature of 50 °C. This excellent performance was ascribed to the fact that h-BN does not react with Li and exhibit high mechanical strength, which allows them to sustain the pressure caused by Li dendrite growth.⁹⁹

However, cracks and holes could be induced on the h-BN during chemical vapor deposition synthesis. To address this issue, in another study, Xie, Jin, *et al.* deposited LiF on h-BN, thereby stitching the fragments of h-BN by LiF and thus completely suppressed dendritic lithium formation owing to the reinforced mechanical stability. The LiF/h-BN coated Li metal electrodes displayed good cycling behavior with a high coulombic efficiency of >95 % after 300 cycles.¹⁰⁰

Moreover, 2D materials can also be used as composites with Li metal for dendrite growth prevention. In a recent study, Liu, Shan, *et al.*, demonstrated the use of 3D crumpled graphene balls (CGBs) as scaffolds to stabilize Li metal anodes. The CGBs were prepared by an aerosol-assisted capillary compression method. The as-obtained CGBs were found to be aggregation resistant, lightweight, lithiophilic, and mechanically robust. The CGBs can easily assemble into a densely packed structure, with abundant evenly distributed, free volume and voids inside and in between the balls at the submicron scale. Consequently, 12 mAh/cm² of Li deposited on a 40 μm thick CGB electrode resulted in a smooth surface and no dendrite formation. The CGBs exhibited a high coulombic efficiency of 97.5 % over 700 cycles.¹⁰¹

Designing 3D current collector based on 2D materials have also shown to effectively suppress dendritic lithium formation. For example, Yang, Guanhui, *et al.* demonstrated the use of a 3D current collector based on graphene anchored on copper foam (GN@Cu foam). The GN@Cu foam lithium electrode displayed a high coulombic efficiency of 98.6 % after 250 cycles in addition to a lithium-dendrite free morphology was revealed. Thus, the GN@Cu foam can serve as a conductive matrix with lithiophilic sites to store Li. In a full cell assembly, with GN@Cu foam Li electrodes as anodes and LiFePO₄ as cathode, a reversible capacity of 154.84 mAh/g was obtained after 100 cycles at 0.2 C.¹⁰²

In addition to graphene and h-BN, MXenes have also been studied as promising 2D materials to suppress lithium dendrite formation. Zhang, Xinyue, *et al.*, reported the use of 3D porous MXene aerogels as scaffolds (Ti_3C_2) for lithium metal anode. A smooth morphology of the MXene aerogel scaffold was achieved without dendrite penetration on the aerogel surface owing to the abundant lithiophilic nucleation sites. Furthermore, the lithium metal anode with the MXene aerogel as a scaffold exhibited a high Coulombic efficiency up to 97.8% at 3 mAh/cm^2 for 30 cycles.¹⁰³

3.3. Electrolytes: Current issues and preventative measures using 2D materials

In conventional LIBs, the electrolyte, corresponding to a solution of a lithium salt (e.g., LiPF_6) in organic solvents such as ethylene carbonate-dimethyl carbonate (EC-DMC), exhibits high flammability and vapor pressure and is, therefore, the most important factor that needs to be controlled to address safety issues.^{104,105} The low stability, high vapor pressure, and flammability of LIB electrolytes, as well as their incompatibility with the environment of conventional electrolyte materials, pose significant hazards. Specifically, electrolyte degradation features the hydrolysis and thermal decomposition of LiPF_6 , which results in the generation of hazardous HF that is also corrosive to the electrodes.^{106,107}

The safety issues of LIBs arising from the use of conventional electrolytes are commonly addressed by the use of additives such as flame retardants,¹⁰⁸ redox shuttles,¹⁰⁹ shutdown additives,¹¹⁰ electrolyte salt stabilizers,¹¹¹ and non-flammable electrolytes such as ionic liquids,^{112–115}

Even though, these safety measures could effectively enhance the safety of the flammable electrolytes they still feel short in some aspects. Flame retardants are limited by their negative effect on batteries electrochemical performance (deteriorating cycling performance and rate capability).¹¹⁶ The low-potential electrochemical system of redox shuttles is a major limitation.¹¹⁷ Moreover, the complete termination of the cell using a shutdown additive makes this type of safety protection receive little interest.¹¹⁸ The poor low-temperature performance of electrolyte stabilizers is limiting their use as a safety measure.¹¹⁹ On the other hand, although ionic liquids are receiving great interest as alternative electrolytes because of their low flammability and low volatility, their ionic conductivities and viscosities still need to be improved.¹²⁰

Moreover, another approach involves the use of alternative electrolytes such as solid electrolytes made from organic polymers (e.g., solid polymers and polymer gels) or inorganic ceramics.¹²¹

Although polymer electrolytes are a good choice for high-temperature applications; however, polymers tend to lose their mechanical strength following long exposure at high temperatures¹²². In addition to the low room temperature conductivity limits its use in high-temperature LIB applications¹²³. On the other hand, among the inorganic ceramics, bentonite clay is commonly used, however; the structure of bentonite clay is made up of exchangeable ions and high charge density in the interlayer space which could hinder lithium ion transport¹²⁴.

As introduced earlier, to address the abovementioned issue, h-BN among the 2D materials was introduced for the first time as electrolyte in LIBs in the study reported by Rodrigues *et al.*, h-BNs intrinsic features which include thermal conductivity, thermal stability, and being chemically inert makes it a good candidate for use as electrolytes in LIBs. Rodrigues *et al.* reported that a ceramic-based electrolyte containing h-BN and an ionic liquid could effectively operate at temperatures of 24–150 °C with an increase in the electrochemical window. LiTFSI/PP13 (ionic liquid)/h-BN, used as an electrolyte in a quasi-solid-state system, was found to be safe up to 150 °C and achieved a specific capacity of 158 mAh/g and a total capacity loss of less than 3 % up to 5 V for 50 cycles.²²

Moreover, h-BN/PAN composite nanofibers have been used as LIB electrolytes, achieving a specific capacity of 144 mAh/g with a capacity retention of 92.4 % after 50 cycles in addition to a high operating voltage of 4.7 V, which was ascribed to the high electrolyte uptake caused by high ionic conductivity (1.0×10^{-3} S/cm). The use of h-BN with polymers enhances thermal conductivity property, and thus, the thermogravimetric analysis showed that the h-BN/PAN composite was stable up to 280 °C. Moreover, h-BN highly contributes to electrolyte stability as well as proton conduction.¹²⁵

The use of h-BN has also been explored in gel polymer electrolytes (GPE) in LIBs for lithium dendrite suppression. Shim, Jimin, *et al.* reported the use of 2D boron nitride nanoflakes (BNNF)

as additives in GPE (G-CFBN). The addition of 0.5 wt% of BNNFs into the GPE resulted in an enhanced ionic conductivity, Li^+ transference number, mechanical modulus, and dendrite-suppressing capability. Due to these unique properties the G-CFBN demonstrated a high specific capacity and excellent rate capability and cycling stability in lithium metal batteries. The G-CFBN in a Li/LFP battery resulted in high reversible capacity of 82 mAh/g for 500 cycles at 10 °C.¹²⁶

In addition to h-BN, a recent study by Pan, Qiwei, et al., involved the use of MXenes as electrolytes for LIBs. MXene ($\text{Ti}_3\text{C}_2\text{T}_x$) was added to a poly(ethylene oxide)/LiTFSI complex ($\text{PEO}_{20}\text{-LiTFSI}$) to prepare MXene-based CPEs (MCPEs). The presence of MXenes enhanced the conductivity with 3.6 wt% of Mxene in CPEs showed a high ionic conductivity of $2.2 \times 10^{-5} \text{ S/m}$ at 28 °C. When 1.5wt% MXene in CPEs was used as the electrolyte in $\text{LiFePO}_4/\text{Li}$ battery, the electrochemical studies showed a high reversible capacity of 140 mAh/g for 50 cycles at a current rate of C/3 and tested at 60 °C. This is ascribed to the well-dispersed 2D Mxene filler in the polymer matrix.¹²⁷

3.4.Separators: Current issues and preventative measures using 2D materials

Separators play a major role in preventing the contact between the positive and negative electrodes that can result in short-circuiting and are commonly represented by polyolefin (polyethylene and polypropylene) microporous membranes in the case of LIBs. In addition, a separator provides a channel for the migration of lithium ions. Although these separators can be engineered to induce thermal shutdown at a certain temperature for safety purposes, they are not able to withstand high temperatures for a long time, which leads to internal electrical short-circuiting.¹²⁸ In addition, the continuous growth of lithium dendrites can pierce the polymer-based separators and also causes the batteries to short circuit.

The use of electrolyte-philic, mechanically and thermally stable material with the polymer-based separators is one solution to overcome the abovementioned issues. Among all the materials, 2D materials such as h-BN have been favored for use in separators owing to the above intrinsic features.

Luo, Wei, *et al.* showed that an h-BN-coated separator can prevent Li dendrite growth and enables stable electrochemical performance with 92 % coulombic efficiency up to 1 V operating voltage

for 100 cycles at 0.5 mA/cm^2 which is ascribed to the more homogeneous thermal distribution of the thermally conductive BN coating.¹²⁹

Recently, Waqas, Muhammad, *et al.*, reported the use of polyethylene-hexagonal boron nitride/poly(vinylidene fluoride-hexafluoropropylene) (PE-BN/PVDF-HFP) bilayer separator. The (PE-BN/PVDF-HFP) bilayer separator was synthesized through a wet chemistry method. The addition of h-BN in PE could effectively suppress dendrite growth due to h-BNs strong adsorption energy with polymers, large interactive surface area, and superior mechanical properties. The bilayer separator owns a high ionic conductivity up to $4.4 \times 10^{-4} \text{ S/cm}$ with a thermal shrinkage of 6.6 % upon annealing at 140°C for 1 h, evidencing the high thermal stability of the separator (Figure 4 (a-d)). Consequently, the PE-BN/PVDF-HFP separator in a lithium iron phosphate/lithium battery exhibited a reversible capacity of 120 mAhg after 500 cycles at 2 C (Figure 4 (e)).¹³⁰

Beyond h-BN, graphene oxide was recently used in separators to stabilize the lithium metal anode. Foroozan, Tara, *et al.*, spray coated graphene oxide nanosheets (GOn) on a glass fiber (GF) separator. SEM images revealed that a dense and uniform deposition of Li was achieved by the presence of GOn with no overgrowth of lithium dendrites following 160 cycles of charge/discharge at 1 mA cm^2 . As a result, enhanced electrochemical performances were obtained for the GOn coated GF separator based cell as a coulombic efficiency of $> 90 \%$ at 1 mA/cm^2 for 30 cycles was achieved.¹³¹

Similarly, in a recent study, Kim, Ju Young, *et al.*, investigated the effect of surface modification of a PE separator using GO. GO-SiO₂ was dip-coated on a PE separator composite layer, and the enhanced electrochemical properties of the GO-SiO₂ coated separator was confirmed in lithium metal secondary batteries. In a full-cell assembly, with LiCoO₂ cathode, lithium metal anode and the GO-SiO₂ modified separator displayed stable cycling performance up to 200 cycles and stable coulombic efficiency was quite stable up to 350 cycles. This enhanced performance is due to GO and SiO₂ nanoparticles ability to effectively suppress lithium dendrite growth.¹³²

Table 2 surveys the use of 2D materials as battery components in LIBs, additionally comparing their performance.

4. Beyond LIBs: 2D materials for high-temperature Li-S batteries

Batteries that are composed of cheap, abundantly available materials safely operate in a wide temperature range while exhibiting increased energy densities are attracting attention as next-generation energy storage systems. To meet the increasing demands for high-performance rechargeable batteries, a number of non-LIB chemistries have been developed. Most importantly, the design and chemistries of such batteries should enable safe operation in a wide temperature range without compromising the electrochemical performance and other factors.¹³³ Li-S batteries are the most promising non-LIB systems, exhibiting gravimetric (2500 Wh/kg) and volumetric (2800 Wh/l) energy densities that are approximately three times higher than those of LIBs.

In terms of composition, Li-S batteries comprise a Li metal anode, an organic liquid electrolyte, and S-based cathode, and their operation is based on a stoichiometric redox reaction between the S cathode (highest theoretical capacity among the cathode materials (1675 mAh/g)) and the Li anode.⁷ Compared to that of LIBs, the active material of Li-S batteries (sulfur) is much more abundant, cheap, lightweight, and environmentally friendly, which makes the latter batteries more economic, lightweight, and safe.¹³⁴

Interestingly, Li-S batteries exhibit an improved safety profile, providing an intrinsic mechanism of overcharge protection based on the formation of a protective layer over Li and additionally exhibiting the advantages of containing a high-flash-point electrolyte and not suffering from Li dendrite growth.^{135,136} The above safety features allow Li-S batteries to be operated in a wide temperature range, as exemplified by the battery developed by OXIS Energy Ltd., which could be safely operated at temperatures from ambient to 140 °C and therefore represented a promising energy storage system particularly suited for oilfield, automotive, and aerospace applications.¹³⁷

Although Li-S batteries seem to overcome the inherent drawbacks of LIBs, they suffer from significant disadvantages that impede commercialization. Both the charge and discharge products, sulfur and lithium sulfide (Li_2S_2 and/or Li_2S), respectively, are insulators (e.g., sulfur has a

conductivity of only 5×10^{-30} S/cm at 25 °C), and their formation, therefore, deteriorates electrode kinetics and cycling capacities. Multiple cycling results in the generation of long-chain lithium polysulfides (Li_2S_x) that dissolve in the electrolyte and can diffuse through the separator to the anode, where they are reduced by Li metal, which results in the loss of sulfur, self-discharge, and rapid capacity decay – a phenomenon known as the polysulfide shuttle.^{138–141,140,141} Full lithiation to Li_2S results in a significant volume expansion of the sulfur cathode (~76%) and therefore leads to electrode damage.⁷

In view of the limitations discussed above, extensive research efforts are currently directed at the development of better Li-S battery cathodes and electrolytes. Importantly, the cathode material should exhibit good sulfur anchoring sites to prevent the dissolution of polysulfides into the electrolyte and the associated shuttle effect caused by polysulfide diffusion from the cathode to the anode and other issues.

4.1.2D materials as Cathodes in Li-S batteries

The use of 2D additives such as graphene can help to overcome the poor electrical conductivity of sulfur and sulfides in Li-S batteries, thereby facilitating electronic transport on the cathode.^{142,143} In addition, volume changes during lithiation/delithiation can also be prevented, which leads to stable cycling performance.¹⁴² Moreover, the increased S-nano additive interface area observed for high-sulfur-affinity nanomaterials such as carbon¹⁴² increases the kinetic barrier for lithium polysulfide loss to the electrolyte.

Graphene and its derivatives have attracted considerable interest as host materials for sulfur in Li-S batteries owing to the fascinating above mentioned properties. To date, various researches were conducted on graphene-based sulfur cathodes. Liu, Feiyan, *et al.* prepared reduced graphene oxide (rGO) encapsulated sulfur through a redox reaction between sodium polysulfide. The sulfur particles were well encapsulated by reduced graphene oxide, and a resistance lower than GO was achieved, and thus the rGO encapsulated sulfur showed a high initial specific capacity of 827 mAh/g with a reversible capacity of 388 mAh/g after 100 cycles at 0.2 C when used as a cathode material for LSBs.¹⁴⁴

For further improvement, graphene and its derivatives were employed by many groups in combination with carbon or polymer to host sulfur cathodes in Li-S batteries.^{145,146} For example, Peng, Huifen, *et al.* prepared a flexible freestanding sulfur/polyacrylonitrile/graphene oxide (S/PAN/GO) via vacuum filtration. GO can effectively anchor polysulfides through the electrostatic interactions between the negatively-charged species and the oxygen functional groups on GO. In addition, the oxygen-containing functional groups on GO could provide active sites for coordination with such as sulfur, and thus improve the electrochemical performance of S/PAN nanocomposite as a cathode composite for Li-S batteries. The battery-based on S/PAN/GO ternary system showed a high specific capacity of 1205 mAh/g after 100 cycles (87 % capacity retention) at 0.5 C, and a high Coulombic efficiency of ~ 96% was obtained as well), indicating that the dissolution of polysulfides and loss of active sulfur was suppressed by the ternary system when compared to S/PAN and S/GO.¹⁴⁷

Inspired by the intrinsic properties of graphene and h-BN, Deng *et al.*, demonstrated the synergistic effect between graphene and BN nanosheets on the adsorption of polysulfides under extreme temperatures. The alternative surface electronic structure graphene-supported BN nanosheets (BN/graphene) composites could lead to unique adsorption characteristics. Compared to pure graphene and h-BN, BN/graphene composites had higher adsorption to polysulfides, and thus a stable capacity with 888 mAh/g at 2 C after 300 cycles with a capacity loss than 14 % at 70 °C was achieved.¹⁴⁸

In addition to graphene and h-BN, extensive researches have been conducted to investigate transition metal chalcogenides¹⁴⁹ including 2D MoS₂ as host materials for sulfur in Li-S batteries. Li, Xueliang, *et al.* reported that the incorporation of MoS₂ in a carbon aerogel composite (coaxial nanocable carbon aerogel (CNCA)) matrix can act as an anchor for soluble polysulfide to due to the copious S atoms on the MoS₂ surface. When formed into a composite with Sulfur, the MoS₂/CNCA/S battery exhibited a high reversible capacity of 1087.83 mAh/g for over 200 cycles at 0.2 C. This high electrochemical performance is because the MoS₂ was evenly embedded in CNCA, in addition, the composite exhibited interconnected, narrow distribution of pore size and high conductivity.¹⁵⁰

It was further reported by Lin, Haibin, *et al.*, that a sulfur-deficient MoS_2 in rGO can act as a catalyst facilitate polysulfide conversion reactions to insoluble products thereby decreasing their accumulation in the sulfur cathode and their loss from the cathode. The MoS_{2-x} /reduced graphene oxide (MoS_{2-x} /rGO) in composite with sulfur exhibited long-term cycling performance with a reversible capacity of 628.2 mAh/g after 600 cycles at 0.5 C. In contrast to, MoS_2 /rGO/S and rGO/S which displayed severe capacity fading after 150 cycles. The high performance could be ascribed to the accelerated polysulfide conversion kinetics, making MoS_2 good catalysis for use in Li-S batteries.¹⁵¹ Later, a similar study was conducted by Wei, Yanju, *et al.*, where MoS_2 was used as a catalyst in MoS_2 /S/rGO composite. The corresponding cathode exhibited long term cyclability with a reversible capacity of 954 mAh/g after 150 cycles at 0.2 C.¹⁵²

Moreover, 2D MoS_2 nanoflakes were also used to encapsulate sulfur, thus limiting polysulfide diffusion as demonstrated by Tang, Wei, *et al.* where MoS_2 flakes self-assemble to form hollow and highly conformal cages to entrap sulfur (Figure 5 (a-g)). The MoS_2 -encapsulated sulfur spheres exhibited outstanding electrochemical performance with a high reversible capacity of 956 mAh/g after 300 cycles at 0.5 C in addition to good cyclability for 1000 cycles at 1 C (Figure 5 (h, i)). Thanks to the high flexibility and strong van der Waal force of MoS_2 which allowed for effective encapsulation of sulfur.¹⁵³

Similarly, Cheng, Songpu, *et al.*, used MoS_2 to encapsulate sulfur in a core-shell structure (MoS_2 @S) with MoS_2 as a shell and sulfur in the core. The MoS_2 @S cathode displayed enhanced electrochemical performance with a reversible capacity of 632 mAh/g after 200 cycles (79 % capacity retention) at current rate of 1C. In contrast to MoS_2 /S which exhibited a lower electrochemical performance of 296.9 mAh/g after 200 cycles (51 % capacity retention) at a current of 1 C was achieved.¹⁵⁴

In addition to MoS_2 , MXenes have been widely investigated as sulfur hosts in Li-S batteries. Bao, Weizhai, *et al.* prepared $\text{Ti}_3\text{C}_2\text{T}_x$ /reduced graphene oxide/sulfur composite through effective two-step liquid phase impregnation method as illustrated in Figure 6 (a), where RGO nanosheets are introduced between the $\text{Ti}_3\text{C}_2\text{T}_x$ sheets to prevent their restacking (Figure 6 (c)). Figure 6 (d) shows the homogenous distribution of sulfur within the $\text{Ti}_3\text{C}_2\text{T}_x$ /reduced graphene oxide heterostructure.

As a result, the $\text{Ti}_3\text{C}_2\text{T}_x$ /reduced graphene oxide/sulfur composite cathodes exhibited a reversible capacity of 878.4 mAh/g after 300 cycles Figure 6 (e). This could be attributed to 3 D layer structure and functional 2 D surfaces of MXene and reduced graphene oxide nanosheets that could effectively trap polysulfides.¹⁵⁵

Recently, sulfur encapsulated by TiO_2 (S@TiO_2) were introduced into Mxene (Ti_2C) layers by Du, Chang, *et al.*. The addition of Mxenes to S@TiO_2 could increase the conductivity and buffer the volumetric expansion of sulfur during charge/discharge processes. Thereby, the $\text{S@TiO}_2/\text{Ti}_2\text{C}$ cathode displayed enhanced electrochemical performance with stable capacities of 464.0 and 227.3 mAh g^{-1} after 200 cycles at high currents of 2 C and 5 C, respectively.¹⁵⁶

Considering the intrinsic features of both MoS_2 and MXenes, Zhang, Yelong, *et al.* reported a nanohybrid composed of MXene/1T-2H MoS_2 -C nanohybrids that could adsorb polysulfides and facilitate the reaction. Consequently, 1 MXene/1T-2H MoS_2 -C-S cathode delivered a high reversible capacity of 799.3 mAh g^{-1} after 300 cycles at 0.5 C.¹⁵⁷

4.2.2D materials as Anodes in Li-S batteries

As discussed previously, 2D materials are promising materials as artificial SEI applications due to their unique features such as high Young's modulus. Thus, graphene and its derivative have already been tested in Li-S battery anodes in reference to dendritic-growth suppression.

Similar to LIBs, the direct protection of lithium metal is one of the approaches to suppress lithium dendritic growth arising from the lithium metal anode in Li-S batteries.

Bobnar, Jernej, *et al.* reported the use of fluorinated reduced graphene oxide interlayer on a lithium metal anode (FGI@Li) as a dendrite suppression layer. They prepared fluorinated reduced graphene oxide (FG), which was then drop cast on a Li-foil surface. An XPS analysis was performed to determine the electrolyte degradation of the lithium surface. Results showed that no increase in electrolyte degradation was observed from the 30th to 100th cycle when FGI@Li was used in contrast to the non-protected lithium which showed an increase in the concentration

of electrolyte degradation products indicating the presence of unstable SEI and decomposition of the electrolyte on the lithium surface.¹⁵⁸

In a recent study, Chen, Yu-Ting, *et al.*, decorated graphene oxide (GO) layers onto Li metal anodes via drop-casting and spray-coating. The dendrite suppression ability of the GO coating was testing in Li|Li symmetrical cell at a current density of 5 mA/cm² for 2000 cycles. No short-circuiting or dramatic change in polarization was observed. This could be ascribed to GOs low nucleation overpotential, and the transformation of GO to reduced GO with cycling enhances its conductivity. In a Li-S battery, GO-decorated Li anodes displayed a reversible capacity of 400 mAh/g over 800 cycles at 1 C.¹⁵⁹

Beyond graphene, 2D MoS₂ has also been used as protective layers for lithium metal anodes in Li-S batteries. Cha, Eunho, *et al.* recently deposited a 10 nm thick layer of MoS₂ on a Li metal surface. A smooth surface with no dendrite growth was observed following 300 cycles of Li deposition/dissolution at a current density of 10 mA/cm², indicating the strong interfacial bonding of 2D MoS₂ to the surface of Li metal. In a full-cell configuration, the Li-S battery with a Li-MoS₂ anode and a CNT-S cathode exhibited a high reversible capacity of 940 mAh/g after 1,200 cycles at 0.5 C.¹⁶⁰

4.1.2D materials as Separators and Interlayers in Li-S batteries

As seen earlier, separators, in LIB batteries are used for two main purposes. One is to prevent a short circuit between the cathode and the anode. The second is to act as a channel for the migration of lithium ions. Besides these two, in Li-S batteries, the separator can also be used to suppress the diffusion of polysulfides. To meet these expectations, various attempts have been made to modify separators and more recently by using an extra layer above the separator called an interlayer.¹⁶¹

Carbons, oxides, and their composites have been used to modify the separators^{162,163} or as an interlayer for better battery performance. However, high mass density loading due to the modification or extra layer could increase the internal resistance and thus lower energy densities of Li-S batteries. Thus, it is essential to develop effective lightweight interlayers.¹⁶⁴

Recently, 2D materials including graphene, MoS₂, MXenes, h-BN due to its ultrathin, lightweight, large surface area, mechanical flexibility, tunable chemistry, have been used for modification of separators or as interlayers in Li-S batteries. To this end, the use of 2D materials in separators and as an interlayer in Li-S batteries has been extensively investigated in the last two years as summarized in Table 3.

This section does not involve 2D materials for electrolytes, as they have not been yet explored as electrolytes in Li-S batteries. Also, the current battery electrolyte of Li-S with 1 M LiTFSI in DIOX: DME (1:1 v/v) (+ LiNO₃), is a safe alternative for conventional LIB electrolytes and to date is the best choice of electrolyte in Li-S batteries even at conditions of severe polysulfide shuttle.

Table 3 presents the major advances of using 2D materials as Li-S components, listing the specific capacities over a number of cycles at a specific current rate.

5. Summary and future outlook

The unique structure and outstanding properties of graphene have inspired the so-called “beyond-graphene” area investigations, which, however, are still in their early phase. To date, most of the properties of beyond-graphene 2D materials have only been probed by theoretical studies, which inspired the development of numerous synthetic approaches to the above materials. 2D materials are viewed as promising components of LIB, exhibiting high surface area to volume ratios and quantum confinement effects and therefore being promising candidates for energy storage applications. Moreover, the synergy between graphene and other 2D materials is of high importance in this area of materials science. This work outlines the contribution of 2D materials to the field of energy storage, specifically to the fabrication of high-temperature Li-based rechargeable batteries, showing that the synthesis of battery component materials for the construction of LIBs with high energy density, long cycle lifetime, and improved safety profile remains a major challenge. Beyond LIBs, rechargeable batteries such as Li-S batteries are viewed as promising systems for complex applications, e.g., those requiring operation under the conditions of high temperature. However, the widespread use of Li-S batteries is hindered by the occurrence of polysulfide dissolution, which results in decreased electrochemical performance and thermal

stability. To solve this and other issues, a deep understanding of the underlying chemistry is required. Finally, we would like to stress that although a number of challenges in the “beyond-LIB” area remain unsolved, significant progress has been recently achieved in mechanistic understanding and choice of suitable materials.

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Competing interests

The authors declare no competing financial interests.

References

1. Chen, J. Recent Progress in Advanced Materials for Lithium Ion Batteries. *Materials* **6**, 156–183 (2013).
2. Thackeray, M. M., Wolverton, C. & Isaacs, E. D. Electrical energy storage for transportation—approaching the limits of, and going beyond, lithium-ion batteries. *Energy Environ. Sci.* **5**, 7854–7863 (2012).
3. Oh, Y., Nam, S., Wi, S., Hong, S. & Park, B. Nanoscale interface control for high-performance Li-ion batteries. *Electron. Mater. Lett.* **8**, 91–105 (2012).
4. Guo, J. & Wang, C. Nanostructured Metal Oxides for Li-Ion Batteries. in *Functional Metal Oxide Nanostructures* 337–363 (Springer, New York, NY, 2012). doi:10.1007/978-1-4419-9931-3_14
5. Janek, J. & Zeier, W. G. A solid future for battery development. *Nature Energy* (2016). doi:10.1038/nenergy.2016.141
6. Manthiram, A. An Outlook on Lithium Ion Battery Technology. *ACS Cent. Sci.* **3**, 1063–1069 (2017).
7. Gu, X., Hencz, L. & Zhang, S. Recent Development of Carbonaceous Materials for Lithium–Sulphur Batteries. *Batteries* **2**, 33 (2016).

8. Fu, R. *et al.* Comparison of Lithium-Ion Anode Materials Using an Experimentally Verified Physics-Based Electrochemical Model. *Energies* **10**, 2174 (2017).
9. Berckmans, G. *et al.* Cost Projection of State of the Art Lithium-Ion Batteries for Electric Vehicles Up to 2030. *Energies* **10**, 1314 (2017).
10. Nitta, N., Wu, F., Lee, J. T. & Yushin, G. Li-ion battery materials: present and future. *Mater. Today* **18**, 252–264 (2015).
11. Zou, P. *et al.* Directing lateral growth of lithium dendrites in micro-compartmented anode arrays for safe lithium metal batteries. *Nat. Commun.* **9**, 464 (2018).
12. Leng, F., Tan, C. M. & Pecht, M. Effect of Temperature on the Aging rate of Li Ion Battery Operating above Room Temperature. *Sci. Rep.* **5**, 12967 (2015).
13. Lin, X., Salari, M., Arava, L. M. R., Ajayan, P. M. & Grinstaff, M. W. High temperature electrical energy storage: advances, challenges, and frontiers. *Chem. Soc. Rev.* **45**, 5848–5887 (2016).
14. Hensley, D., Milewits, M. & Zhang, W. The Evolution of Oilfield Batteries. *Oilfield Rev.* 42–57 (1998).
15. Willgoos, K. Ultracapacitors in downhole technology for Oil and Gas. (2017). Available at: <https://fastcapsystems.com/blog.html#!/posts/Ultracapacitors-in-downhole-technology-for-Oil-and-Gas/4>. (Accessed: 27th October 2017)
16. Brambilla, N. & Martini, F. Ultra-High Temperature (250 C+) and Wide Operating Temperature Range Ultracapacitor Enabling Downhole Power Source for Geothermal Exploration. *GRC Trans.* **39**, 1065–1068 (2015).
17. Coordinator. High Temperature, High Shock Rechargeable Battery Design (CHiTS – Cell for High Temperature/Shock). *GCCIR*
18. Caja, J., Dunstan, T. D. J. & Caja, M. Rechargeable, Lithium-Ion Molten Salt Battery for High Temperature Applications. *ECS Trans.* **50**, 3–11 (2013).
19. Janakiraman, U. *et al.* Development of a High Temperature CFx Battery for Downhole Applications.
20. Oil and Gas. EaglePicher Technologies, LLC, 2013. *Eagle Picher* Available at: www.eaglepicher.com/r-n-d/new-product-development/down-hole-batteries.

21. Fripp, M. L. *et al.* Development of a High-Temperature Rechargeable Battery for Downhole Use in Petroleum Industry. in (Offshore Technology Conference, 2008). doi:10.4043/19621-MS
22. Rodrigues, M.-T. F. *et al.* Hexagonal Boron Nitride-Based Electrolyte Composite for Li-Ion Battery Operation from Room Temperature to 150 °C. *Adv. Energy Mater.* **6**, n/a-n/a (2016).
23. Williams, M. Battery components can take the heat. (2016). Available at: <http://news.rice.edu/2016/04/11/battery-components-can-take-the-heat/>. (Accessed: 27th October 2017)
24. Zhou, D., Cui, Y. & Han, B. Graphene-based hybrid materials and their applications in energy storage and conversion. *Chin. Sci. Bull.* **57**, 2983–2994 (2012).
25. Guo, Y.-G., Hu, J.-S. & Wan, L.-J. Nanostructured Materials for Electrochemical Energy Conversion and Storage Devices. *Adv. Mater.* **20**, 2878–2887 (2008).
26. Jiang, C., Hosono, E. & Zhou, H. Nanomaterials for lithium ion batteries. *Nano Today* **1**, 28–33 (2006).
27. Lu, J. *et al.* The role of nanotechnology in the development of battery materials for electric vehicles. *Nat. Nanotechnol.* **11**, n nano.2016.207 (2016).
28. Mahmood, N. & Hou, Y. Electrode Nanostructures in Lithium-Based Batteries. *Adv. Sci.* **1**, (2014).
29. Bruce, P. G., Scrosati, B. & Tarascon, J.-M. Nanomaterials for Rechargeable Lithium Batteries. *Angew. Chem. Int. Ed.* **47**, 2930–2946 (2008).
30. Sun, Y., Liu, N. & Cui, Y. Promises and challenges of nanomaterials for lithium-based rechargeable batteries. *Nat. Energy* **1**, nenergy201671 (2016).
31. Song, X., Wang, X., Sun, Z., Zhang, P. & Lian, G. Recent Developments in Silicon Anode Materials for High Performance Lithium-Ion Batteries. *Sigma-Aldrich* Available at: <https://www.sigmaaldrich.com/technical-documents/articles/materials-science/recent-developments-in-silicon-anode-materials.html>. (Accessed: 6th November 2017)
32. Wang, B. Silicon put in graphene cages for a battery anode able to hold ten times more charge. *NextBigFuture.com* (2016). Available at: <https://www.nextbigfuture.com/2016/02/silicon-put-in-graphene-cages-for.html>. (Accessed: 6th November 2017)
33. Salisbury, D. Quantum dots made from fool's gold boost battery performance. *Vanderbilt University* (2015). Available at: <https://news.vanderbilt.edu/2015/11/11/quantum-dots->

- made-from-fool%e2%80%99s-gold-boost-battery-performance/. (Accessed: 6th November 2017)
34. Mas-Ballesté, R., Gómez-Navarro, C., Gómez-Herrero, J. & Zamora, F. 2D materials: to graphene and beyond. *Nanoscale* **3**, 20–30 (2011).
35. Gupta, A., Sakthivel, T. & Seal, S. Recent development in 2D materials beyond graphene. *Prog. Mater. Sci.* **73**, 44–126 (2015).
36. Luo, J., Gao, J., Wang, A. & Huang, J. Bulk Nanostructured Materials Based on Two-Dimensional Building Blocks: A Roadmap. *ACS Nano* **9**, 9432–9436 (2015).
37. Sahoo, R., Pal, A. & Pal, T. 2D materials for renewable energy storage devices: Outlook and challenges. *Chem. Commun.* **52**, 13528–13542 (2016).
38. Miró, P., Audiffred, M. & Heine, T. An atlas of two-dimensional materials. *Chem. Soc. Rev.* **43**, 6537–6554 (2014).
39. Wang, X., Weng, Q., Yang, Y., Bando, Y. & Golberg, D. Hybrid two-dimensional materials in rechargeable battery applications and their microscopic mechanisms. *Chem. Soc. Rev.* **45**, 4042–4073 (2016).
40. Chen, K.-S., Balla, I., Luu, N. S. & Hersam, M. C. Emerging Opportunities for Two-Dimensional Materials in Lithium-Ion Batteries. *ACS Energy Lett.* **2**, 2026–2034 (2017).
41. Zhang, C. *et al.* 2D Materials for Lithium/Sodium Metal Anodes. *Adv. Energy Mater.* **8**, 1802833 (2018).
42. Qin, G. & Hu, M. Diverse Thermal Transport Properties of Two-Dimensional Materials: A Comparative Review. (2016). doi:10.5772/64298
43. Trivedi, S., Srivastava, A. & Kurchania, R. Silicene and Germanene: A First Principle Study of Electronic Structure and Effect of Hydrogenation-Passivation. *J. Comput. Theor. Nanosci.* **11**, 781–788 (2014).
44. Balendhran, S., Walia, S., Nili, H., Sriram, S. & Bhaskaran, M. Elemental Analogues of Graphene: Silicene, Germanene, Stanene, and Phosphorene. *Small* **11**, 640–652 (2015).
45. Kulish, V. V., Malyi, O. I., Persson, C. & Wu, P. Adsorption of metal adatoms on single-layer phosphorene. *Phys. Chem. Chem. Phys.* **17**, 992–1000 (2014).
46. Wang, J., Li, G. & Li, L. Synthesis Strategies about 2D Materials. (2016). doi:10.5772/63918
47. Rani, P. & Jindal, V. K. Designing band gap of graphene by B and N dopant atoms. *RSC Adv.* **3**, 802–812 (2012).

48. Zhu, Y. *et al.* Graphene and Graphene Oxide: Synthesis, Properties, and Applications. *Adv. Mater.* **22**, 3906–3924 (2010).
49. Biró, L. P., Nemes-Incze, P. & Lambin, P. Graphene: nanoscale processing and recent applications. *Nanoscale* **4**, 1824–1839 (2012).
50. Dodson, B. New materials mimic electronic properties of graphene in 3D. *New Atlas* (2014). Available at: <https://newatlas.com/3d-graphene-semimetal-electronics/30495/>. (Accessed: 6th November 2017)
51. Rao, C. N. R., Sood, A. K., Voggu, R. & Subrahmanyam, K. S. Some Novel Attributes of Graphene. *J. Phys. Chem. Lett.* **1**, 572–580 (2010).
52. Novoselov, K. S. *et al.* Electronic properties of graphene. *Phys. Status Solidi B* **244**, 4106–4111 (2007).
53. Geim, A. K. & Novoselov, K. S. The rise of graphene. *Nat. Mater.* **6**, nmat1849 (2007).
54. Rao, C. N. R., Biswas, K., Subrahmanyam, K. S. & Govindaraj, A. Graphene, the new nanocarbon. *J. Mater. Chem.* **19**, 2457–2469 (2009).
55. Nan, H. Y. *et al.* The thermal stability of graphene in air investigated by Raman spectroscopy. *J. Raman Spectrosc.* **44**, 1018–1021 (2013).
56. Han, W.-Q. Anisotropic Hexagonal Boron Nitride Nanomaterials: Synthesis and Applications. in *Nanotechnologies for the Life Sciences* (Wiley-VCH Verlag GmbH & Co. KGaA, 2007). doi:10.1002/9783527610419.ntls0161
57. Weng, Q., Wang, X., Wang, X., Bando, Y. & Golberg, D. Functionalized hexagonal boron nitride nanomaterials: emerging properties and applications. *Chem. Soc. Rev.* **45**, 3989–4012 (2016).
58. Gao, R. *et al.* High-Yield Synthesis of Boron Nitride Nanosheets with Strong Ultraviolet Cathodoluminescence Emission. *J. Phys. Chem. C* **113**, 15160–15165 (2009).
59. Wang, Y., Shi, Z. & Yin, J. Boron nitride nanosheets: large-scale exfoliation in methanesulfonic acid and their composites with polybenzimidazole. *J. Mater. Chem.* **21**, 11371–11377 (2011).
60. Wang, L., Sun, C., Xu, L. & Qian, Y. Convenient synthesis and applications of gram scale boron nitride nanosheets. *Catal. Sci. Technol.* **1**, 1119–1123 (2011).
61. Lin, Y. & Connell, J. W. Advances in 2D boron nitride nanostructures: nanosheets, nanoribbons, nanomeshes, and hybrids with graphene. *Nanoscale* **4**, 6908–6939 (2012).

62. Lin, Y., Williams, T. V. & Connell, J. W. Soluble, Exfoliated Hexagonal Boron Nitride Nanosheets. *J. Phys. Chem. Lett.* **1**, 277–283 (2010).
63. Pakdel, A., Zhi, C., Bando, Y. & Golberg, D. Low-dimensional boron nitride nanomaterials. *Mater. Today* **15**, 256–265 (2012).
64. Aldalbahi, A., Zhou, A. F. & Feng, P. Variations in Crystalline Structures and Electrical Properties of Single Crystalline Boron Nitride Nanosheets. *Sci. Rep.* **5**, srep16703 (2015).
65. Zhou, H. *et al.* High thermal conductivity of suspended few-layer hexagonal boron nitride sheets. *Nano Res.* **7**, 1232–1240 (2014).
66. Song, L. *et al.* Large scale growth and characterization of atomic hexagonal boron nitride layers. *Nano Lett.* **10**, 3209–3215 (2010).
67. Pop, E., Varshney, V. & Roy, A. K. Thermal properties of graphene: Fundamentals and applications. *MRS Bull.* **37**, 1273–1281 (2012).
68. Jiao, Y. *et al.* Metallic MoS₂ for High Performance Energy Storage and Energy Conversion. *Small* **14**, 1800640 (2018).
69. Wang, T., Chen, S., Pang, H., Xue, H. & Yu, Y. MoS₂-Based Nanocomposites for Electrochemical Energy Storage. *Adv. Sci.* **4**, (2016).
70. Zhang, P., Wang, F., Yu, M., Zhuang, X. & Feng, X. Two-dimensional materials for miniaturized energy storage devices: from individual devices to smart integrated systems. *Chem. Soc. Rev.* **47**, 7426–7451 (2018).
71. Han, S. A., Bhatia, R. & Kim, S.-W. Synthesis, properties and potential applications of two-dimensional transition metal dichalcogenides. *Nano Converg.* **2**, 17 (2015).
72. Liu, D.-S. *et al.* Tailored performance of layered transition metal dichalcogenides via integration with low dimensional nanostructures. *RSC Adv.* **7**, 11987–11997 (2017).
73. Pumera, M., Sofer, Z. & Ambrosi, A. Layered transition metal dichalcogenides for electrochemical energy generation and storage. *J. Mater. Chem. A* **2**, 8981–8987 (2014).
74. Emerging Energy Applications of Two-Dimensional Layered Materials. *Can. Chem. Trans.* 118–157 (2015). doi:10.13179/canchemtrans.2015.03.02.0174
75. Pang, J. *et al.* Applications of 2D MXenes in energy conversion and storage systems. *Chem. Soc. Rev.* **48**, 72–133 (2019).
76. Tang, H. *et al.* MXene–2D layered electrode materials for energy storage. *Prog. Nat. Sci. Mater. Int.* **28**, 133–147 (2018).

77. Sun, S., Liao, C., Hafez, A. M., Zhu, H. & Wu, S. Two-dimensional MXenes for energy storage. *Chem. Eng. J.* **338**, 27–45 (2018).
78. Suna, S., Liaoa, C., Hafez, A. M., Zhuc, H. & Wua, S. Two-dimensional MXenes for energy storage. in (2018).
79. Chen, R., Zhao, T., Zhang, X., Li, L. & Wu, F. Advanced cathode materials for lithium-ion batteries using nanoarchitectonics. *Nanoscale Horiz.* **1**, 423–444 (2016).
80. Aravindan, V., Gnanaraj, J., Lee, Y.-S. & Madhavi, S. LiMnPO₄ – A next generation cathode material for lithium-ion batteries. *J. Mater. Chem. A* **1**, 3518–3539 (2013).
81. Wang, Q. *et al.* Thermal runaway caused fire and explosion of lithium ion battery. *J. Power Sources* **208**, 210–224 (2012).
82. Balakrishnan, P. G., Ramesh, R. & Prem Kumar, T. Safety mechanisms in lithium-ion batteries. *J. Power Sources* **155**, 401–414 (2006).
83. Wen, J., Yu, Y. & Chen, C. A Review on Lithium-Ion Batteries Safety Issues: Existing Problems and Possible Solutions. *Mater. Express* **2**, 197–212 (2012).
84. Mun, J. *et al.* New dry carbon nanotube coating of over-lithiated layered oxide cathode for lithium ion batteries. *J. Mater. Chem. A* **2**, 19670–19677 (2014).
85. Zhou, X., Wang, F., Zhu, Y. & Liu, Z. Graphene modified LiFePO₄ cathode materials for high power lithium ion batteries. *J. Mater. Chem.* **21**, 3353–3358 (2011).
86. Lung-Hao Hu, B., Wu, F.-Y., Lin, C.-T., Khlobystov, A. N. & Li, L.-J. Graphene-modified LiFePO₄ cathode for lithium ion battery beyond theoretical capacity. *Nat. Commun.* **4**, 1687 (2013).
87. Fang, X., Ge, M., Rong, J. & Zhou, C. Graphene-oxide-coated LiNi_{0.5}Mn_{1.5}O₄ as high voltage cathode for lithium ion batteries with high energy density and long cycle life. *J. Mater. Chem. A* **1**, 4083–4088 (2013).
88. Venkateswara Rao, C., Leela Mohana Reddy, A., Ishikawa, Y. & Ajayan, P. M. LiNi_{1/3}Co_{1/3}Mn_{1/3}O₂–Graphene Composite as a Promising Cathode for Lithium-Ion Batteries. *ACS Appl. Mater. Interfaces* **3**, 2966–2972 (2011).
89. Wang, X. *et al.* Graphene-decorated carbon-coated LiFePO₄ nanospheres as a high-performance cathode material for lithium-ion batteries. *Carbon* **127**, 149–157 (2018).
90. Son, I. H. *et al.* Graphene balls for lithium rechargeable batteries with fast charging and high volumetric energy densities. *Nat. Commun.* **8**, 1561 (2017).

91. School of Automobile and Traffic Engineering, Wuhan University of Science and Technology, Wuhan 430081, China & Ma, Y. Graphene Oxide Modified LiNi_{1/3}Co_{1/3}Mn_{1/3}O₂ as Cathode Material for Lithium Ion Batteries and Its Electrochemical Performances. *Int. J. Electrochem. Sci.* 10440–10453 (2018). doi:10.20964/2018.11.17
92. Shim, J.-H., Kim, Y.-M., Park, M., Kim, J. & Lee, S. Reduced Graphene Oxide-Wrapped Nickel-Rich Cathode Materials for Lithium Ion Batteries. *ACS Appl. Mater. Interfaces* **9**, 18720–18729 (2017).
93. Fu, Y. *et al.* LiFePO₄-Graphene Composites as High-Performance Cathodes for Lithium-Ion Batteries: The Impact of Size and Morphology of Graphene. *Materials* **12**, 842 (2019).
94. Stephenson, T., Li, Z., Olsen, B. & Mitlin, D. Lithium ion battery applications of molybdenum disulfide (MoS₂) nanocomposites. *Energy Environ. Sci.* **7**, 209–231 (2013).
95. Wang, C.-H. *et al.* Titanium Carbide (MXene) as a Current Collector for Lithium-Ion Batteries. *ACS Omega* **3**, 12489–12494 (2018).
96. Lin, D., Liu, Y. & Cui, Y. Reviving the lithium metal anode for high-energy batteries. *Nat. Nanotechnol.* **12**, 194–206 (2017).
97. Tao, T. *et al.* Anode Improvement in Rechargeable Lithium–Sulfur Batteries. *Adv. Mater.* n/a–n/a doi:10.1002/adma.201700542
98. Lin, D., Liu, Y., Pei, A. & Cui, Y. Nanoscale perspective: Materials designs and understandings in lithium metal anodes. *Nano Res.* **10**, 4003–4026 (2017).
99. Yan, K. *et al.* Ultrathin Two-Dimensional Atomic Crystals as Stable Interfacial Layer for Improvement of Lithium Metal Anode. *Nano Lett.* **14**, 6016–6022 (2014).
100. Xie, J. *et al.* Stitching h-BN by atomic layer deposition of LiF as a stable interface for lithium metal anode. *Sci. Adv.* **3**, eaao3170 (2017).
101. Liu, S. *et al.* Crumpled Graphene Balls Stabilized Dendrite-free Lithium Metal Anodes. *Joule* **2**, 184–193 (2018).
102. Yang, G. *et al.* Graphene anchored on Cu foam as a lithiophilic 3D current collector for a stable and dendrite-free lithium metal anode. *J. Mater. Chem. A* **6**, 9899–9905 (2018).
103. Zhang, X. *et al.* MXene Aerogel Scaffolds for High-Rate Lithium Metal Anodes. *Angew. Chem. Int. Ed.* **57**, 15028–15033 (2018).

104. Scrosati, B. & Garche, J. Lithium batteries: Status, prospects and future. *J. Power Sources* **195**, 2419–2430 (2010).
105. Armand, M. & Tarascon, J.-M. Building better batteries. *Nature* **451**, 652–657 (2008).
106. Terborg, L. *et al.* Investigation of thermal aging and hydrolysis mechanisms in commercial lithium ion battery electrolyte. *J. Power Sources* **242**, 832–837 (2013).
107. Campion, C. L., Li, W. & Lucht, B. L. Thermal Decomposition of LiPF₆-Based Electrolytes for Lithium-Ion Batteries. *J. Electrochem. Soc.* **152**, A2327–A2334 (2005).
108. Hyung, Y. E., Vissers, D. R. & Amine, K. Flame-retardant additives for lithium-ion batteries. *J. Power Sources* **119–121**, 383–387 (2003).
109. Wen, J., Yu, Y. & Chen, C. A Review on Lithium-Ion Batteries Safety Issues: Existing Problems and Possible Solutions. *Mater. Express* **2**, 197–212 (2012).
110. Xia, L., Wang, D., Yang, H., Cao, Y. & Ai, X. An electrolyte additive for thermal shutdown protection of Li-ion batteries. *Electrochem. Commun.* **25**, 98–100 (2012).
111. Xu, C. *et al.* LiTDI: A Highly Efficient Additive for Electrolyte Stabilization in Lithium-Ion Batteries. *Chem. Mater.* **29**, 2254–2263 (2017).
112. Navarra, M. A. Ionic liquids as safe electrolyte components for Li-metal and Li-ion batteries. *MRS Bull.* **38**, 548–553 (2013).
113. Scrosati, B., Hassoun, J. & Sun, Y.-K. Lithium-ion batteries. A look into the future. *Energy Environ. Sci.* **4**, 3287–3295 (2011).
114. Armand, M., Endres, F., MacFarlane, D. R., Ohno, H. & Scrosati, B. Ionic-liquid materials for the electrochemical challenges of the future. *Nat. Mater.* **8**, nmat2448 (2009).
115. Lewandowski, A. & Świdarska-Mocek, A. Ionic liquids as electrolytes for Li-ion batteries—An overview of electrochemical studies. *J. Power Sources* **194**, 601–609 (2009).
116. Hess, S., Wohlfahrt-Mehrens, M. & Wachtler, M. Flammability of Li-Ion Battery Electrolytes: Flash Point and Self-Extinguishing Time Measurements. *J. Electrochem. Soc.* **162**, A3084–A3097 (2015).
117. Zhang, L., Zhang, Z. & Amine, K. Redox Shuttle Additives for Lithium-Ion Battery. in *Lithium Ion Batteries - New Developments* (ed. Belharouak, I.) (InTech, 2012). doi:10.5772/26572
118. Zhang, Z. & Zhang, S. S. *Rechargeable Batteries: Materials, Technologies and New Trends*. (Springer, 2015).

119. Täubert, C., Fleischhammer, M., Wohlfahrt-Mehrens, M., Wietelmann, U. & Buhrmester, T. LiBOB as Electrolyte Salt or Additive for Lithium-Ion Batteries Based on $\text{LiNi}_{0.8}\text{Co}_{0.15}\text{Al}_{0.05}\text{O}_2$ /Graphite. *J. Electrochem. Soc.* **157**, A721 (2010).
120. Balbuena, P. B. Electrolyte materials - Issues and challenges. *AIP Conf. Proc.* **1597**, 82–97 (2014).
121. Fergus, J. W. Ceramic and polymeric solid electrolytes for lithium-ion batteries. *J. Power Sources* **195**, 4554–4569 (2010).
122. Manuel Stephan, A. & Nahm, K. S. Review on composite polymer electrolytes for lithium batteries. *Polymer* **47**, 5952–5964 (2006).
123. Zhang, R. *et al.* Advanced Micro/Nanostructures for Lithium Metal Anodes. *Adv. Sci.* **4**, (2017).
124. Fu, X. *et al.* Inorganic and organic hybrid solid electrolytes for lithium-ion batteries. *CrystEngComm* **18**, 4236–4258 (2016).
125. Aydın, H., Çelik, S. Ü. & Bozkurt, A. Electrolyte loaded hexagonal boron nitride/polyacrylonitrile nanofibers for lithium ion battery application. *Solid State Ion.* **309**, 71–76 (2017).
126. Shim, J. *et al.* 2D boron nitride nanoflakes as a multifunctional additive in gel polymer electrolytes for safe, long cycle life and high rate lithium metal batteries. *Energy Environ. Sci.* **10**, 1911–1916 (2017).
127. Pan, Q. *et al.* 2D MXene-containing polymer electrolytes for all-solid-state lithium metal batteries. *Nanoscale Adv.* **1**, 395–402 (2019).
128. Yan, X. *et al.* Polyimide binder by combining with polyimide separator for enhancing the electrochemical performance of lithium ion batteries. *Electrochimica Acta* **216**, 1–7 (2016).
129. Luo, W. *et al.* A Thermally Conductive Separator for Stable Li Metal Anodes. *Nano Lett.* **15**, 6149–6154 (2015).
130. Waqas, M. *et al.* High-Performance PE-BN/PVDF-HFP Bilayer Separator for Lithium-Ion Batteries. *Adv. Mater. Interfaces* **6**, 1801330 (2019).
131. Foroozan, T. *et al.* Synergistic Effect of Graphene Oxide for Impeding the Dendritic Plating of Li. *Adv. Funct. Mater.* **28**, 1705917 (2018).
132. Kim, J. Y. *et al.* Graphene Oxide Induced Surface Modification for Functional Separators in Lithium Secondary Batteries. *Sci. Rep.* **9**, 2464 (2019).

133. Chen, T. *et al.* Self-Templated Formation of Interlaced Carbon Nanotubes Threaded Hollow Co₃S₄ Nanoboxes for High-Rate and Heat-Resistant Lithium–Sulfur Batteries. *J. Am. Chem. Soc.* **139**, 12710–12715 (2017).
134. Conder, J. *et al.* Direct observation of lithium polysulfides in lithium–sulfur batteries using *operando* X-ray diffraction. *Nat. Energy* **2**, nenergy201769 (2017).
135. Kolosnitsyn, V. S. & Karaseva, E. V. Lithium-sulfur batteries: Problems and solutions. *Russ. J. Electrochem.* **44**, 506–509 (2008).
136. Fotouhi, A. *et al.* Lithium #8211;Sulfur Cell Equivalent Circuit Network Model Parameterization and Sensitivity Analysis. *IEEE Trans. Veh. Technol.* **66**, 7711–7721 (2017).
137. OXIS Jump-Starts First Commercialization Of Lithium-Sulfur Batteries. Available at: <http://www.hybridcars.com/oxis-jump-starts-first-commercialization-of-lithium-sulfur-batteries/>. (Accessed: 8th May 2018)
138. Pang, Q., Liang, X., Kwok, C. Y. & Nazar, L. F. Advances in lithium–sulfur batteries based on multifunctional cathodes and electrolytes. *Nat. Energy* **1**, nenergy2016132 (2016).
139. Zhang, J., Yang, C.-P., Yin, Y.-X., Wan, L.-J. & Guo, Y.-G. Sulfur Encapsulated in Graphitic Carbon Nanocages for High-Rate and Long-Cycle Lithium–Sulfur Batteries. *Adv. Mater.* **28**, 9539–9544 (2016).
140. Fang, R. *et al.* More Reliable Lithium-Sulfur Batteries: Status, Solutions and Prospects. *Adv. Mater.* n/a-n/a (2017). doi:10.1002/adma.201606823
141. Pang, Q., Liang, X., Kwok, C. Y., Kulisch, J. & Nazar, L. F. A Comprehensive Approach toward Stable Lithium–Sulfur Batteries with High Volumetric Energy Density. *Adv. Energy Mater.* **7**, n/a-n/a (2017).
142. Ma, L., Hendrickson, K. E., Wei, S. & Archer, L. A. Nanomaterials: Science and applications in the lithium–sulfur battery. *Nano Today* **10**, 315–338 (2015).
143. Li, B., Xu, H., Ma, Y. & Yang, S. Harnessing the unique properties of 2D materials for advanced lithium–sulfur batteries. *Nanoscale Horiz.* **4**, 77–98 (2018).
144. Liu, F. *et al.* Reduced graphene oxide encapsulated sulfur spheres for the lithium-sulfur battery cathode. *Results Phys.* **7**, 250–255 (2017).
145. Li, H. *et al.* Three-Dimensionally Hierarchical Graphene Based Aerogel Encapsulated Sulfur as Cathode for Lithium/Sulfur Batteries. *Nanomater. Basel Switz.* **8**, (2018).

146. Kim, J., Kang, Y., Song, S.-W. & Suk, J. Freestanding sulfur-graphene oxide/carbon composite paper as a stable cathode for high performance lithium-sulfur batteries. *Electrochimica Acta* **299**, 27–33 (2019).
147. Peng, H. *et al.* Synthesis of a Flexible Freestanding Sulfur/Polyacrylonitrile/Graphene Oxide as the Cathode for Lithium/Sulfur Batteries. *Polymers* **10**, 399 (2018).
148. Deng, D. R. *et al.* Enhanced Adsorptions to Polysulfides on Graphene-Supported BN Nanosheets with Excellent Li–S Battery Performance in a Wide Temperature Range. *ACS Nano* **12**, 11120–11129 (2018).
149. Babu, G., Masurkar, N., Al Salem, H. & Arava, L. M. R. Transition Metal Dichalcogenide Atomic Layers for Lithium Polysulfides Electrocatalysis. *J. Am. Chem. Soc.* **139**, 171–178 (2017).
150. Li, X., Zhao, K., Zhang, L., Ding, Z. & Hu, K. MoS₂-decorated coaxial nanocable carbon aerogel composites as cathode materials for high performance lithium-sulfur batteries. *J. Alloys Compd.* **692**, 40–48 (2017).
151. Lin, H. *et al.* Electrocatalysis of polysulfide conversion by sulfur-deficient MoS₂ nanoflakes for lithium–sulfur batteries. *Energy Environ. Sci.* **10**, 1476–1486 (2017).
152. Wei, Y. *et al.* Sulfur film sandwiched between few-layered MoS₂ electrocatalysts and conductive reduced graphene oxide as a robust cathode for advanced lithium–sulfur batteries. *J. Mater. Chem. A* **6**, 5899–5909 (2018).
153. Tang, W. *et al.* In Situ Observation and Electrochemical Study of Encapsulated Sulfur Nanoparticles by MoS₂ Flakes. *J. Am. Chem. Soc.* **139**, 10133–10141 (2017).
154. Cheng, S., Xia, X., Liu, H. & Chen, Y. Core-shell structured MoS₂@S spherical cathode with improved electrochemical performance for lithium-sulfur batteries. *J. Mater. Sci. Technol.* **34**, 1912–1918 (2018).
155. Bao, W. *et al.* Confined Sulfur in 3 D MXene/Reduced Graphene Oxide Hybrid Nanosheets for Lithium–Sulfur Battery. *Chem. – Eur. J.* **23**, 12613–12619 (2017).
156. Du, C. *et al.* Embedding S@TiO₂ nanospheres into MXene layers as high rate cyclability cathodes for lithium-sulfur batteries. *Electrochimica Acta* **295**, 1067–1074 (2019).
157. Zhang, Y. *et al.* Rational Design of MXene/1T-2H MoS₂-C Nanohybrids for High-Performance Lithium–Sulfur Batteries. *Adv. Funct. Mater.* **28**, 1707578 (2018).

158. Bobnar, J. *et al.* Fluorinated reduced graphene oxide as a protective layer on the metallic lithium for application in the high energy batteries. *Sci. Rep.* **8**, 5819 (2018).
159. Chen, Y.-T. *et al.* Mitigating Metal Dendrite Formation in Lithium–Sulfur Batteries via Morphology-Tunable Graphene Oxide Interfaces. *ACS Appl. Mater. Interfaces* **11**, 2060–2070 (2019).
160. Cha, E. *et al.* 2D MoS₂ as an efficient protective layer for lithium metal anodes in high-performance Li-S batteries. *Nat. Nanotechnol.* **13**, 337–344 (2018).
161. Yuan, K. *et al.* Insight into the Function Mechanism of the Carbon Interlayer in Lithium–Sulfur Batteries. *J. Electrochem. Soc.* **165**, A1880–A1885 (2018).
162. Kiai, M. S. & Kizil, H. Functionalized Double Side Coated Separator for Lithium-Sulfur Batteries with Enhanced Cycle Life. *J. Electrochem. Soc.* **165**, A3574–A3581 (2018).
163. He, Y., Qiao, Y. & Zhou, H. Recent advances in functional modification of separators in lithium–sulfur batteries. *Dalton Trans.* **47**, 6881–6887 (2018).
164. Fan, L. *et al.* Interlayer Material Selection for Lithium-Sulfur Batteries. *Joule* **3**, 361–386 (2019).
165. Tang, W. *et al.* High-Performance Solid Polymer Electrolytes Filled with Vertically Aligned 2D Materials. *Adv. Funct. Mater.* **0**, 1900648
166. Qu, L. *et al.* Enhanced Cycling Performance for Lithium–Sulfur Batteries by a Laminated 2D g-C₃N₄/Graphene Cathode Interlayer. *ChemSusChem* **12**, 213–223 (2019).
167. Li, Z. MnO₂-graphene nanosheets wrapped mesoporous carbon/sulfur composite for lithium-sulfur batteries. *R. Soc. Open Sci.* **5**, 171824–171824 (2018).
168. Kong, L. *et al.* Porphyrin-Derived Graphene-Based Nanosheets Enabling Strong Polysulfide Chemisorption and Rapid Kinetics in Lithium–Sulfur Batteries. *Adv. Energy Mater.* **8**, 1800849 (2018).
169. Zhu, X. *et al.* In Situ Assembly of 2D Conductive Vanadium Disulfide with Graphene as a High-Sulfur-Loading Host for Lithium–Sulfur Batteries. *Adv. Energy Mater.* **8**, 1800201 (2018).
170. Deng, D.-R. *et al.* Co₄N Nanosheet Assembled Mesoporous Sphere as a Matrix for Ultrahigh Sulfur Content Lithium–Sulfur Batteries. *ACS Nano* **11**, 6031–6039 (2017).

171. Majumder, S., Shao, M., Deng, Y. & Chen, G. Two Dimensional WS₂/C Nanosheets as a Polysulfides Immobilizer for High Performance Lithium-Sulfur Batteries. *J. Electrochem. Soc.* **166**, A5386–A5395 (2019).
172. Li, M. *et al.* Ultrathin SnS₂ nanosheets as robust polysulfides immobilizers for high performance lithium-sulfur batteries. *Mater. Res. Bull.* **96**, 509–515 (2017).
173. Sun, Z. *et al.* Conductive porous vanadium nitride/graphene composite as chemical anchor of polysulfides for lithium-sulfur batteries. *Nat. Commun.* **8**, 14627 (2017).
174. Ye, C. *et al.* 2D MoN-VN Heterostructure To Regulate Polysulfides for Highly Efficient Lithium-Sulfur Batteries. *Angew. Chem. Int. Ed.* **57**, 16703–16707 (2018).
175. Song, J. *et al.* Rational design of free-standing 3D porous MXene/rGO hybrid aerogels as polysulfide reservoirs for high-energy lithium-sulfur batteries. *J. Mater. Chem. A* **7**, 6507–6513 (2019).
176. Bao, W. *et al.* Facile Synthesis of Crumpled Nitrogen-Doped MXene Nanosheets as a New Sulfur Host for Lithium-Sulfur Batteries. *Adv. Energy Mater.* **8**, 1702485 (2018).
177. Zhao, Q., Zhu, Q., Miao, J., Zhang, P. & Xu, B. 2D MXene nanosheets enable small-sulfur electrode to be flexible for lithium-sulfur batteries. *Nanoscale* (2019). doi:10.1039/C8NR09653H
178. Dong, Y. *et al.* All-MXene-Based Integrated Electrode Constructed by Ti₃C₂ Nanoribbon Framework Host and Nanosheet Interlayer for High-Energy-Density Li-S Batteries. *ACS Nano* (2018). doi:10.1021/acsnano.7b07672
179. Gao, X.-T. *et al.* Ultrathin MXene Nanosheets Decorated with TiO₂ Quantum Dots as an Efficient Sulfur Host toward Fast and Stable Li-S Batteries. *Small* **14**, 1802443 (2018).
180. Zhang, H. *et al.* Self-Assembled 3D MnO₂ Nanosheets@Delaminated-Ti₃C₂ Aerogel as Sulfur Host for Lithium-Sulfur Battery Cathodes. *ACS Appl. Energy Mater.* (2018). doi:10.1021/acsaem.8b01765
181. Patil, S. B. *et al.* Exfoliated 2D Lepidocrocite Titanium Oxide Nanosheets for High Sulfur Content Cathodes with Highly Stable Li-S Battery Performance. *ACS Energy Lett.* **3**, 412–419 (2018).
182. Li, B. *et al.* Flexible Ti₃C₂ MXene-lithium film with lamellar structure for ultrastable metallic lithium anodes. *Nano Energy* **39**, 654–661 (2017).

183. Zhang, L. *et al.* Dual-Functional Graphene Carbon as Polysulfide Trapper for High-Performance Lithium Sulfur Batteries. *ACS Appl. Mater. Interfaces* **10**, 5594–5602 (2018).
184. Lei, T. *et al.* Inhibiting Polysulfide Shuttling with a Graphene Composite Separator for Highly Robust Lithium-Sulfur Batteries. *Joule* **2**, 2091–2104 (2018).
185. Wang, S. *et al.* Two-Dimensional CeO₂/RGO Composite-Modified Separator for Lithium/Sulfur Batteries. *Nanoscale Res. Lett.* **13**, 377 (2018).
186. Shi, N. *et al.* Insight into different-microstructured ZnO/graphene-functionalized separators affecting the performance of lithium–sulfur batteries. *J. Mater. Chem. A* **7**, 4009–4018 (2019).
187. Kim, P. J. H. *et al.* Synergistic protective effect of a BN-carbon separator for highly stable lithium sulfur batteries. *NPG Asia Mater.* **9**, am201751 (2017).
188. Babu, G., Sawas, A., Thangavel, N. K. & Arava, L. M. R. Two-Dimensional Material-Reinforced Separator for Li–Sulfur Battery. *J. Phys. Chem. C* **122**, 10765–10772 (2018).
189. Fan, Y. *et al.* Repelling Polysulfide Ions by Boron Nitride Nanosheet Coated Separators in Lithium–Sulfur Batteries. *ACS Appl. Energy Mater.* (2019). doi:10.1021/acsaem.8b02205
190. Li, C., Gao, K. & Zhang, Z. Graphitic carbon nitride as polysulfide anchor and barrier for improved lithium–sulfur batteries. *Nanotechnology* **29**, 465401 (2018).
191. Ghazi, Z. A. *et al.* MoS₂/Celgard Separator as Efficient Polysulfide Barrier for Long-Life Lithium–Sulfur Batteries. *Adv. Mater.* **29**, 1606817 (2017).
192. Jeong, Y. C. *et al.* Rational design of exfoliated 1T MoS₂@CNT-based bifunctional separators for lithium sulfur batteries. *J. Mater. Chem. A* **5**, 23909–23918 (2017).
193. Han, P., Chung, S.-H. & Manthiram, A. Thin-Layered Molybdenum Disulfide Nanoparticles as an Effective Polysulfide Mediator in Lithium–Sulfur Batteries. *ACS Appl. Mater. Interfaces* **10**, 23122–23130 (2018).
194. Yang, S. *et al.* A 3D MoS₂/Graphene Microsphere Coated Separator for Excellent Performance Li-S Batteries. *Materials* **11**, (2018).
195. Chen, G., Song, X., Wang, S., Chen, X. & Wang, H. Two-dimensional molybdenum nitride nanosheets modified Celgard separator with multifunction for LiS batteries. *J. Power Sources* **408**, 58–64 (2018).
196. Yin, L., Xu, G., Nie, P., Dou, H. & Zhang, X. MXene debris modified eggshell membrane as separator for high-performance lithium-sulfur batteries. *Chem. Eng. J.* **352**, 695–703 (2018).

197. Jiao, L. *et al.* Capture and Catalytic Conversion of Polysulfides by In Situ Built TiO₂-MXene Heterostructures for Lithium–Sulfur Batteries. *Adv. Energy Mater.* **0**, 1900219
198. Fan, Y. *et al.* Functionalized Boron Nitride Nanosheets/Graphene Interlayer for Fast and Long-Life Lithium–Sulfur Batteries. *Adv. Energy Mater.* **7**, 1602380 (2017).
199. Yao, C. *et al.* Self-assembled layer-by-layer partially reduced graphene oxide–sulfur composites as lithium–sulfur battery cathodes. *RSC Adv.* **8**, 3443–3452 (2018).
200. Yin, L. *et al.* A functional interlayer as a polysulfides blocking layer for high-performance lithium–sulfur batteries. *New J. Chem.* **42**, 1431–1436 (2018).
201. Hu, N. *et al.* SnO₂/Reduced Graphene Oxide Interlayer Mitigating the Shuttle Effect of Li–S Batteries. *ACS Appl. Mater. Interfaces* **10**, 18665–18674 (2018).
202. Liu, Y. *et al.* Fe₃O₄-Decorated Porous Graphene Interlayer for High-Performance Lithium–Sulfur Batteries. *ACS Appl. Mater. Interfaces* **10**, 26264–26273 (2018).
203. Shi, H., Dong, Y., Zhou, F., Chen, J. & Wu, Z.-S. 2D hybrid interlayer of electrochemically exfoliated graphene and Co(OH)₂ nanosheet as a bi-functionalized polysulfide barrier for high-performance lithium–sulfur batteries. *J. Phys. Energy* **1**, 015002 (2018).
204. Qin, J.-L. *et al.* Graphene-based Fe-coordinated framework porphyrin as an interlayer for lithium–sulfur batteries. *Mater. Chem. Front.* **3**, 615–619 (2019).
205. Xiao, Z., Yang, Z., Zhou, L., Zhang, L. & Wang, R. Highly Conductive Porous Transition Metal Dichalcogenides via Water Steam Etching for High-Performance Lithium–Sulfur Batteries. *ACS Appl. Mater. Interfaces* **9**, 18845–18855 (2017).
206. Guo, P. *et al.* Dual functional MoS₂/graphene interlayer as an efficient polysulfide barrier for advanced lithium-sulfur batteries. *Electrochimica Acta* **256**, 28–36 (2017).
207. Tan, L., Li, X., Wang, Z., Guo, H. & Wang, J. Lightweight Reduced Graphene Oxide@MoS₂ Interlayer as Polysulfide Barrier for High-Performance Lithium–Sulfur Batteries. *ACS Appl. Mater. Interfaces* **10**, 3707–3713 (2018).
208. Yan, L. *et al.* Enhanced performance of lithium-sulfur batteries with an ultrathin and lightweight MoS₂/carbon nanotube interlayer. *J. Power Sources* **389**, 169–177 (2018).
209. Tang, H. *et al.* In Situ Formed Protective Barrier Enabled by Sulfur@Titanium Carbide (MXene) Ink for Achieving High-Capacity, Long Lifetime Li-S Batteries. *Adv. Sci.* **5**, 1800502 (2018).

210. "Renata Batteries Silver Oxide Coin Cells - High Drain 1.55 V Renata No. 357 Data Sheet." [Online]. Available: <http://www.renata.com/industrial-products/silver-oxide-coin-cells/>. [Accessed: 28-Oct-2017].
211. "Varta Microbattery Primary Alkaline Manganese Cells (Cylindrical) Type 4122 Data Sheet." [Online]. Available: https://products.varta-microbattery.com/applications/mb_data/documents/data_sheets/DS4122.pdf. [Accessed: 27-Oct-2017].
212. "Saft LO/G Cylindrical Primary Lithium Cells based on Lithium Sulfur Dioxide," *Saft Batteries*, 19-May-2017. [Online]. Available: <https://www.saftbatteries.com/products-solutions/products/lo-g>. [Accessed: 28-Oct-2017].
213. "Varta Microbattery Primary Lithium Cells Lithium Manganese Dioxide - Sales Program and Technical Handbook - Cylindrical Cell CRAA Type 6117." [Online]. Available: https://www.varta-microbattery.com/wp-content/uploads/2017/08/HANDBOOK_Primary_Lithium_Cells_en.pdf. [Accessed: 27-Oct-2017].
214. "EaglePicher Technologies, LLC- LiCFx-MnO₂ Hybrid LCF-133 Primary D Cell Data Sheet." [Online]. Available: <https://www.eaglepicher.com/technology/battery-chemistries/lithium-carbon-monofluoride-cfx-and-licfx-mno2-hybrid/>. [Accessed: 28-Oct-2017].
215. "Saft Batteries - Lithium Thionyl Chloride- LS, LSH cylindrical primary lithium cells," *Saft Batteries*, 19-May-2017. [Online]. Available: <https://www.saftbatteries.com/products-solutions/products/ls-lsh>. [Accessed: 28-Oct-2017].
216. "EaglePicher Technologies LLC - LCF-137 Non-Rechargeable Li/CFx Hybrid Data Sheet." [Online]. Available: <https://www.eaglepicher.com/sites/default/files/EP%20LCF%20137%20Preliminary.pdf>. [Accessed: 27-Oct-2017].
217. "Power-Sonic Corporation - NiCd & NiMH Batteries and Chargers Specification and Summary Sheets." [Online]. Available: http://www.power-sonic.com/nickel_cadmium_n_nimh.php. [Accessed: 28-Oct-2017].
218. "VARTA Microbattery -Powerful High Temperature NiMH Batteries - Press Release Electronica 2008." [Online]. Available: <https://products.varta->

- microbattery.com//applications/mb_data/documents/product_information/PI20081111_Elec
tronica_Ni-MH_High_Temp_en.pdf. [Accessed: 27-Oct-2017].
219. “NiMH Batteries (Nickel Metal Hydride): High Temperature & Long Life Type | Panasonic
Industrial Devices.” [Online]. Available:
[https://na.industrial.panasonic.com/products/batteries/rechargeable-batteries/nickel-metal-
hydride/series/high-temperature-long-life-type/CS4395](https://na.industrial.panasonic.com/products/batteries/rechargeable-batteries/nickel-metal-hydride/series/high-temperature-long-life-type/CS4395). [Accessed: 28-Oct-2017].
220. “EaglePicher Technologies LLC - Silver Zinc Battery GAP-4470 Data Sheet.” [Online].
Available: <https://www.eaglepicher.com/sites/default/files/4470.PDF>. [Accessed: 27-Oct-
2017].
221. “EaglePicher Technologies LLC Lead Acid Battery Carefree® CFR-6V58-S9 Data Sheet.”
[Online]. Available: <https://www.eaglepicher.com/sites/default/files/cfr6v58.s9.pdf>.
[Accessed: 27-Oct-2017].
222. “Saft Batteries - Li Ion - MP, small VL,” *Saft Batteries*, 19-May-2017. [Online]. Available:
<https://www.saftbatteries.com/products-solutions/products/mp-small-vl>. [Accessed: 28-Oct-
2017].
223. “Renata Batteries - Rechargeable Lithium Ion Polymer Battery Pack 3.7V Data Sheet for
ICP401230UPR.” [Online]. Available:
[http://www.renata.com/fileadmin/downloads/productsheets/lithium_polymer/ICP401230UP
R.pdf](http://www.renata.com/fileadmin/downloads/productsheets/lithium_polymer/ICP401230UPR.pdf). [Accessed: 28-Oct-2017].
224. “Seiko Instruments Inc. Micro Battery Product Catalogue 2016 - MS Lithium Rechargeable
Battery (MS-T series Wider Temperature Range) Product MS920T/MS621T.” [Online].
Available: http://www.sii.co.jp/en/me/files/2016/02/English_Micro-Battery-2016.pdf.
[Accessed: 27-Oct-2017].
225. H. Kim, J. T. Lee, and G. Yushin, “High temperature stabilization of lithium–sulfur cells with
carbon nanotube current collector,” *J. Power Sources*, vol. 226, no. Supplement C, pp. 256–
265, Mar. 2013.
226. “Oxis Energy Li-S Technology,” *Oxis Energy*. [Online]. Available:
<https://oxisenergy.com/technology/>. [Accessed: 24-Nov-2017].
227. “EaglePicher Technologies LLC - MAR-9522 Lithium Iron Phosphate (LFP) Chemistry
Rechargeable Data Sheet.” [Online]. Available:

- <https://www.eaglepicher.com/sites/default/files/EP%20MAR%209522%20LITHIUM%20ION%20DATA%20SHEET.PDF>. [Accessed: 27-Oct-2017].
228. “Front Edge Technology, Inc - Technical Data of NanoEnergy® Thin Film Lithium Battery.” [Online]. Available: <http://www.frontedgetechnology.com/tech.htm>. [Accessed: 28-Oct-2017].
229. “Excellatron - Advantage of Excellatron’s Thin Film Batteries.” [Online]. Available: <https://www.excellatron.com/advantage.htm>. [Accessed: 28-Oct-2017].
230. M.-T. F. Rodrigues *et al.*, “A materials perspective on Li-ion batteries at extreme temperatures,” *Nat. Energy*, vol. 2, no. 8, p. nenergy2017108, Jul. 2017.
231. K. B. Hueso, M. Armand, and T. Rojo, “High temperature sodium batteries: status, challenges and future trends,” *Energy Environ. Sci.*, vol. 6, no. 3, pp. 734–749, Feb. 2013.
232. “Sodium Sulfur (NAS) Batteries.” [Online]. Available: <http://energystorage.org/energy-storage/technologies/sodium-sulfur-nas-batteries>. [Accessed: 28-Oct-2017].
233. “European Association for Storage of Energy - Sodium Sulfur Battery.” [Online]. Available: http://ease-storage.eu/wp-content/uploads/2016/03/EASE_TD_NaS.pdf. [Accessed: 28-Oct-2017].
234. “European Association for Storage of Energy - Sodium Nickel Chloride Battery.” .
235. N. Stauffer, “A battery made of molten metals,” *MIT News*, 12-Jan-2016. [Online]. Available: <http://news.mit.edu/2016/battery-molten-metals-0112>. [Accessed: 28-Oct-2017].

Figure legends

Figure 1: Classification of high-temperature batteries (high operating temperatures are defined as those exceeding 60 °C).

Figure 2: SEM images of (a) NCM and (b) rGO-NCM of fresh (uncycled) samples. (c, d) Enlarged images of (a) and (b), respectively. HRTEM images of (e) NCM and (f) rGO-NCM. (g) Schematic illustration of rGO wrapped NCMs and (h) cyclic performance of NCMs versus rGO wrapped NCMs.⁹⁰

Figure 3: Representative (a) cross-section and (b) top-view scanning electron microscope (SEM) images of mossy dendrites grown on planar Cu substrate after depositing 0.75 mAh/cm² of Li. Dendrite-free high Amount of Li grown on top of 40 µm thick Crumpled Graphene Ball (GCB) of

electrodes at (c) 6 mAhr/cm² and (d) 12 mAhr/cm². (e) Coulombic efficiency of CGB electrodes and bare Cu electrodes with Li deposition amount of 0.5 mAhr/cm² at a current rate of 0.5 mA/cm².⁹⁹

Figure 4: (a) Thermal shrinkage graph of Celgard 2325 and PE-BN/PVDF-HFP separators at different temperatures for 1 h, (b) TGA curves of Celgard 2325 and PE-BN/PVDF-HFP separators, (c) thermal shrinkage photographs of the Celgard 2325 and PE-BN/PVDF-HFP separators before and after annealing at 140 °C for 1 h, (d) the combustion test photographs of Celgard 2325 and PE-BN/PVDF-HFP separators with 40 μm thickness, and (e) cycle performance of LFP/Li cells based on PE-BN/PVDF-HFP and Celgard 2325 separators at 2 C after 500 cycles.¹²⁸

Figure 5: Hermetic encapsulation of sulfur spheres by flexible MoS₂ flakes. (a) STEM image of typical MoS₂ flakes and (b) atomic STEM images to show the crystalline structure of MoS₂ flakes. (c) STEM images of hollow MoS₂ cages upon removal of sulfur particles. (d) Schematic of (i) the synthesis process of MoS₂-encapsulated hollow sulfur sphere, (ii) effective soluble lithium polysulfides entrapment and structural integrity of MoS₂-encapsulated hollow sulfur sphere upon lithiation/delithiation. (e) STEM image and (f, g) high-resolution TEM images of MoS₂-encapsulated hollow sulfur sphere. Inset of (e) indicates the line scan EDX signal of sulfur. (h) Long-term cycle capability of MoS₂-encapsulated hollow sulfur sphere, rGO mixed with sulfur spheres and hollow sulfur sphere at 0.5C. (g) Ultra-long-term cyclability of MoS₂-encapsulated hollow sulfur sphere at 1C.¹⁵²

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Figures

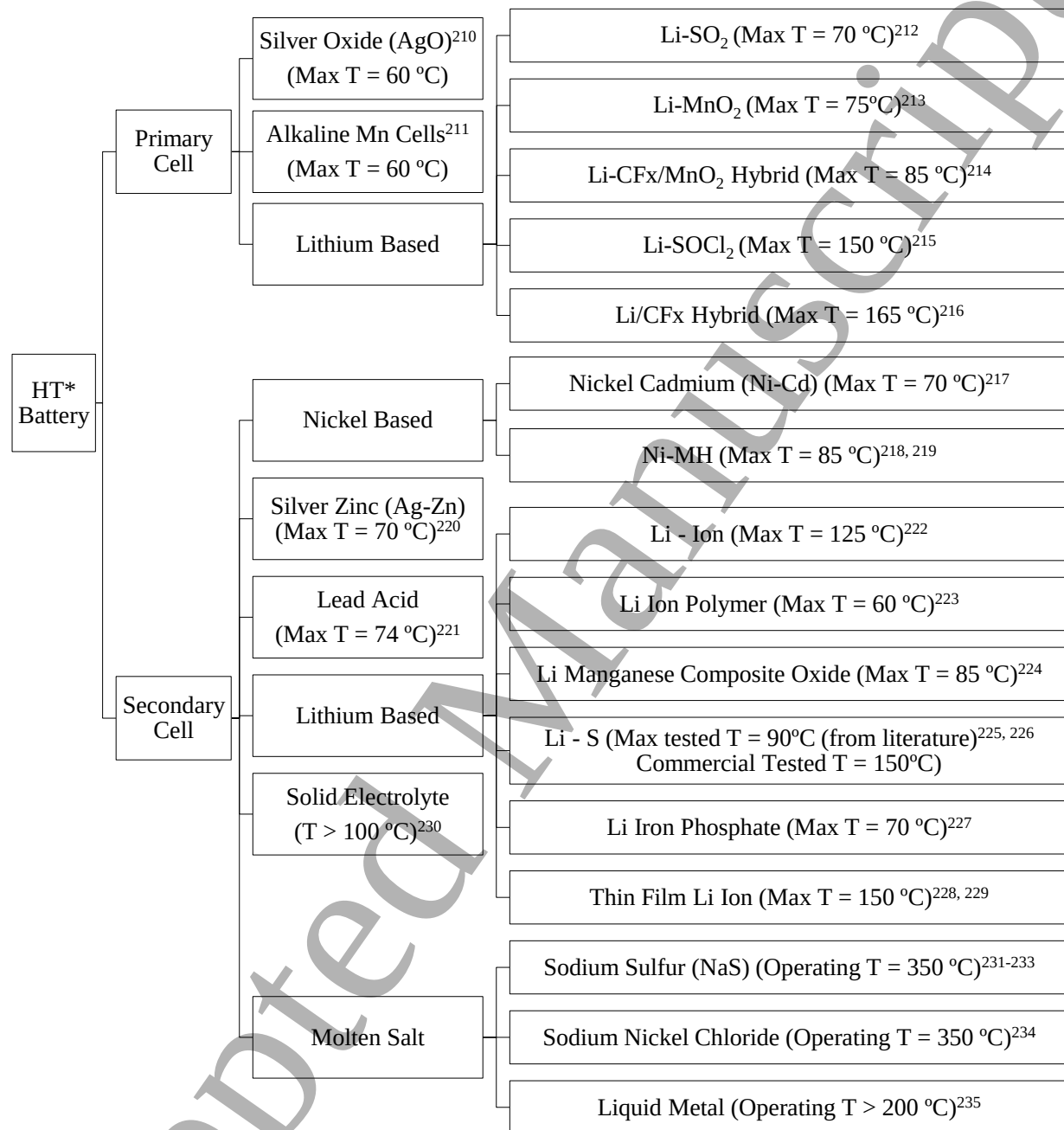


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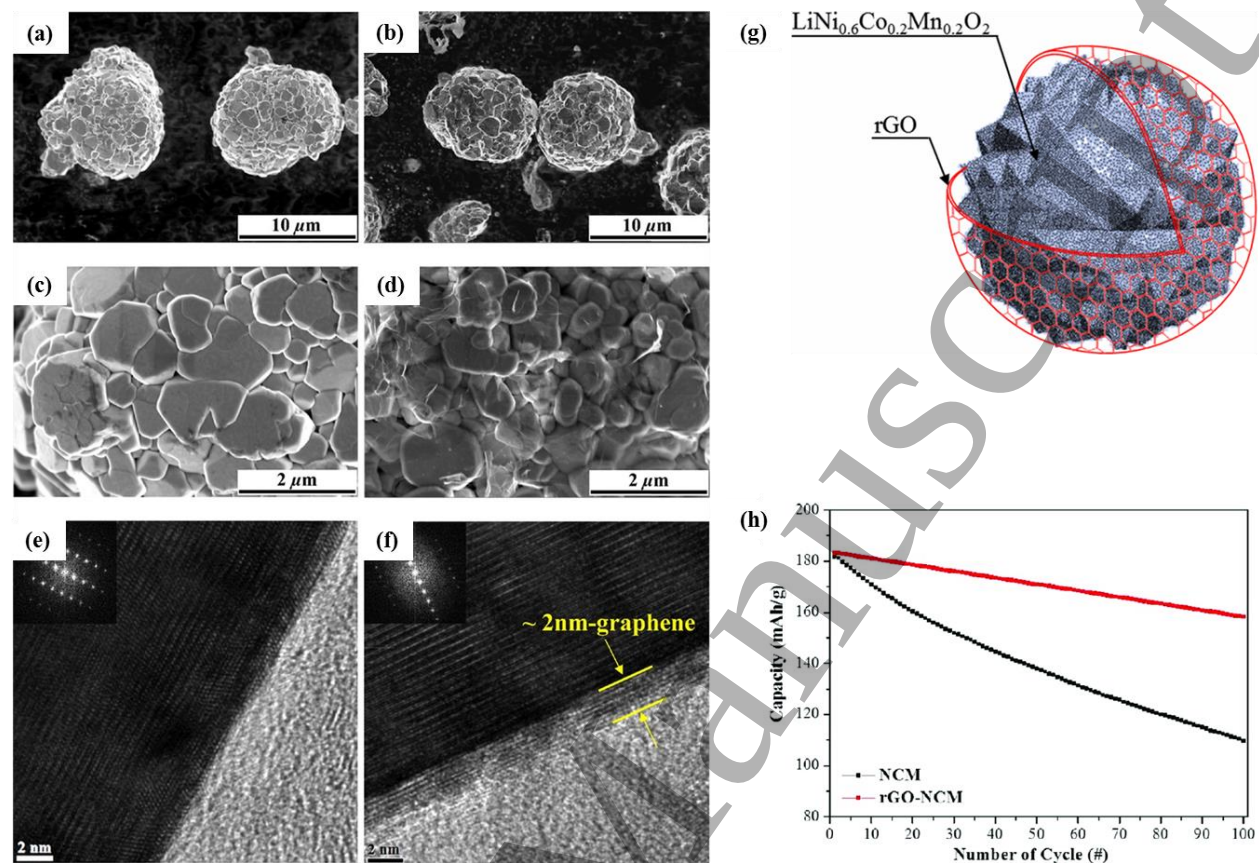


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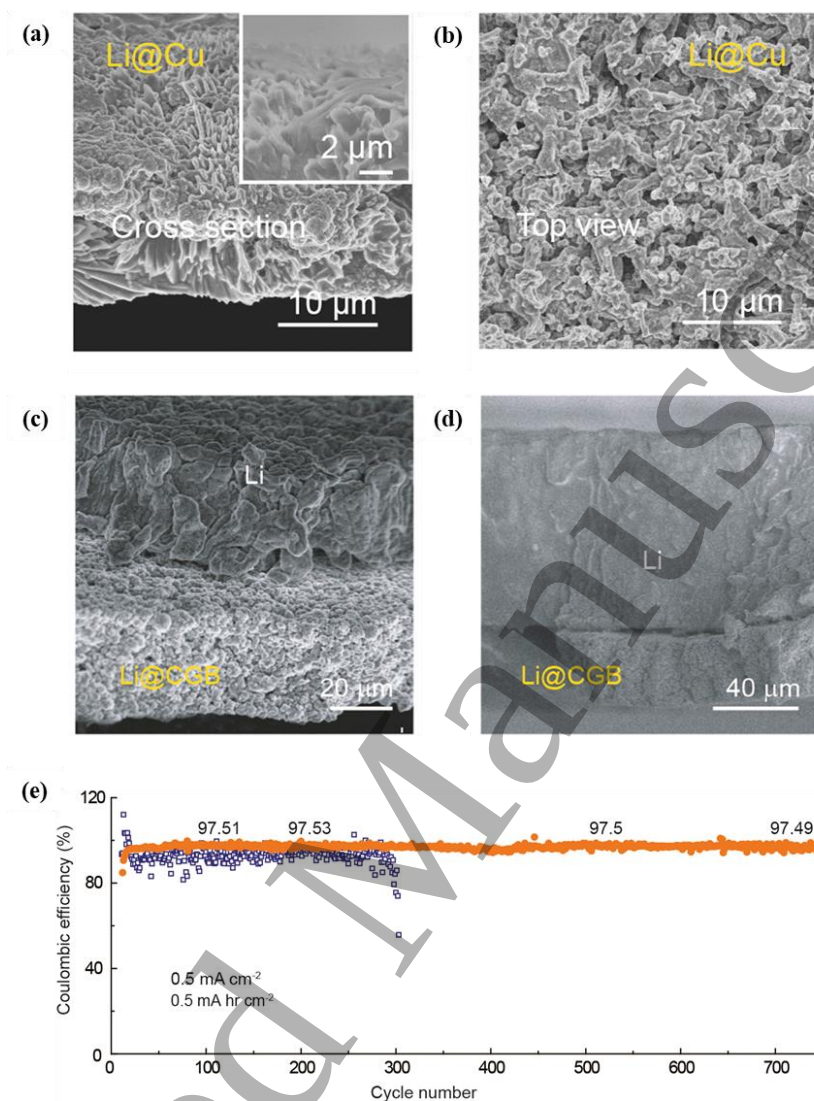


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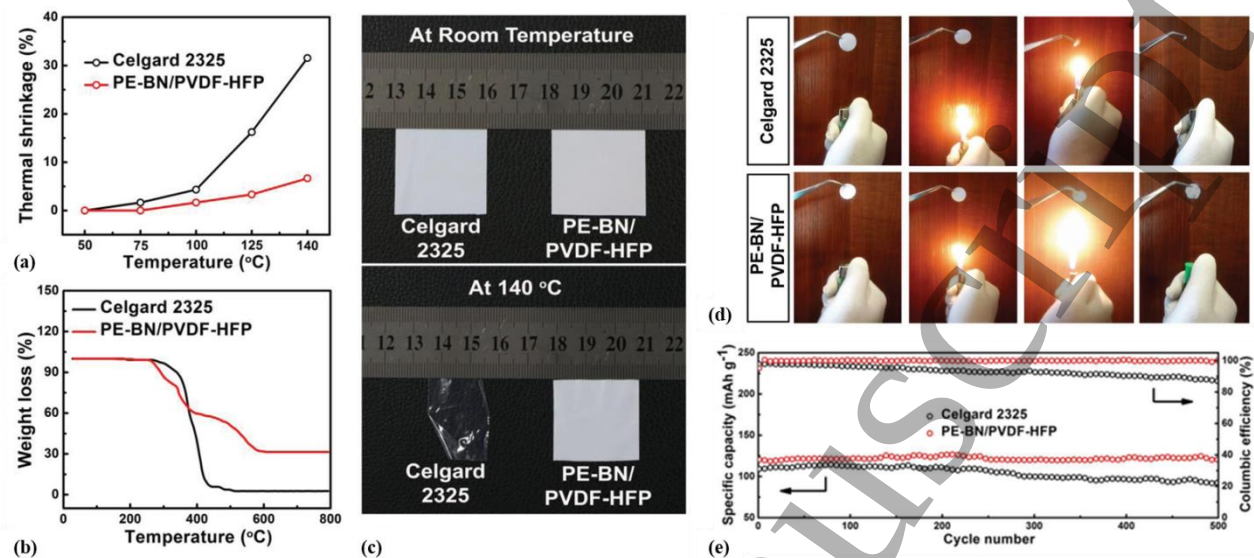


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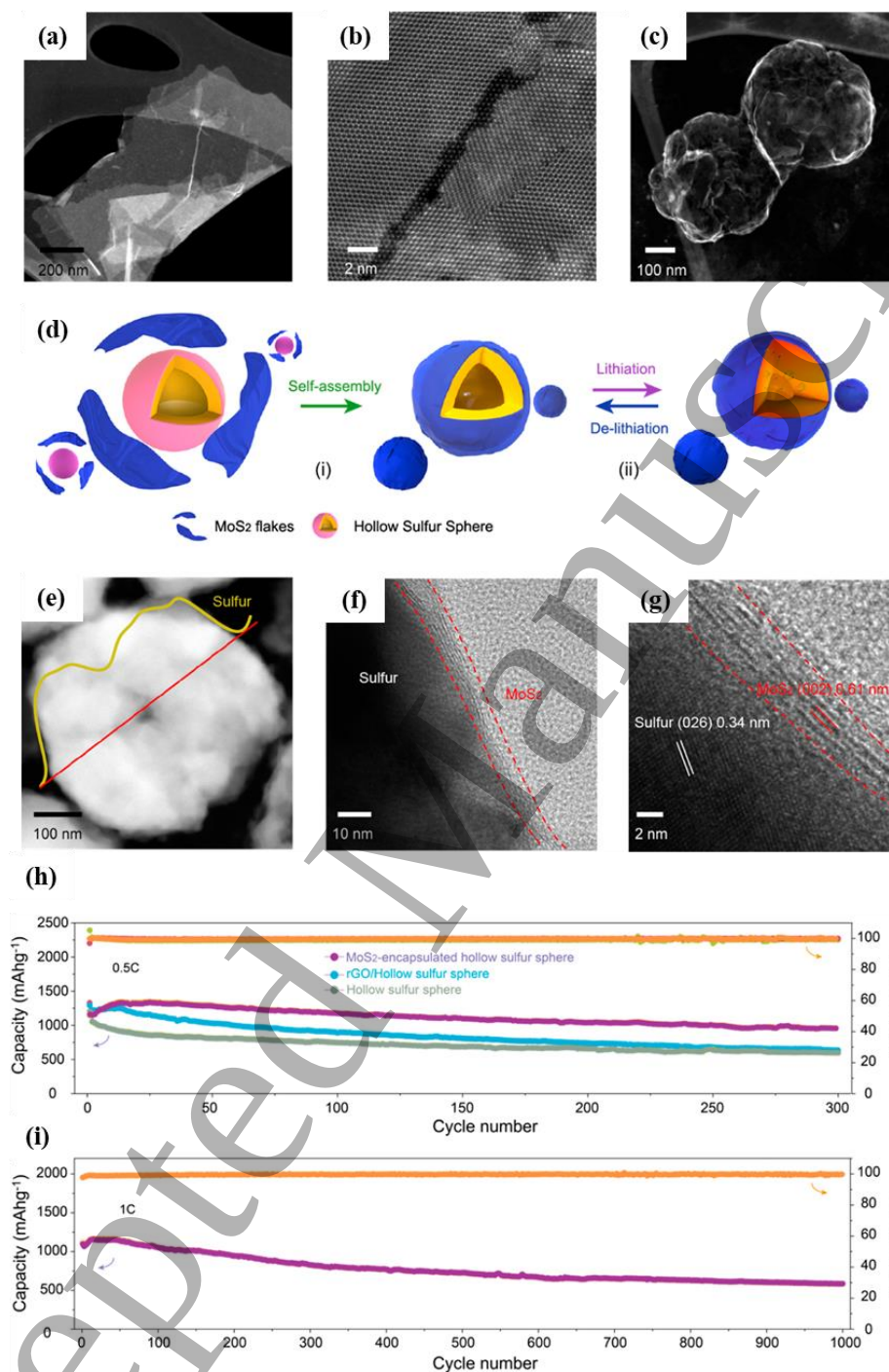


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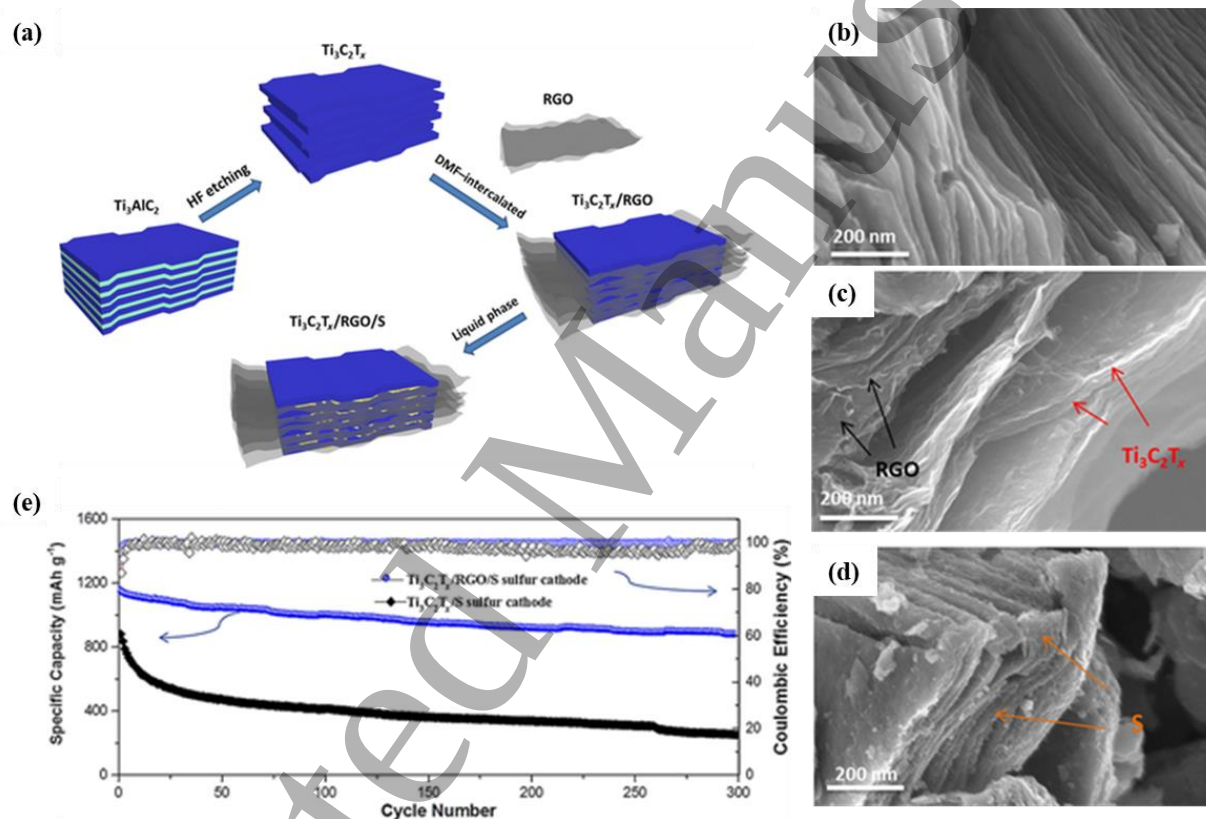


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Tables

Table 1: Conventional LIBs: Cell component requirements, the problem, and the development of new materials.

Cell component	Requirements	Problem	Development	New materials
Cathodes: LiCoO ₂ , LiMn ₂ O ₄ , LiFePO ₄	High capacity, low reactivity with electrolyte, low self-discharge and cyclic performance etc.	No candidate materials with high capacity. Low voltage window. Improved safety.	Design materials with high capacity. Higher potential window. Coating to prevent degradation.	Development of olivine, fluorides etc. Using metal oxides and carbon.
Anodes: Lithium, Silicon, graphite etc.	Li adsorption capability, low reactivity with electrolyte, capacity and cyclic performance etc.	There are candidates with high capacity but short cycle lives. Improved safety.	Transition metal oxides. Coating to prevent lithium dendrite formation.	2D materials such as graphene and h-BN.
Electrolytes: Lithium salts mixed with	Chemical inertness, thermal stability, low viscosity, safe etc.	Low voltage window.	Higher potential window.	Ionic liquids. Ceramics or polymers.

organic solvents.		Improved safety.	Non-flammable. Solid electrolytes.	
Separators: PP, PE, PMMA, PVdF etc.	Porous, prevents short circuit, thin layer and ionic conduction. Chemical, thermal, electrochemical, mechanical, dimensional stability and shutdown at high temperatures.	Low voltage window. Improved safety. Low voltage window. Improved safety.	Higher potential window. High melting point materials.	Ceramics such as h-BN/polymer composites.

Table 2: 2D materials as LIB components.

Battery component	Material	Reversible capacity (mAh/g)	Cycle number	Current rate	Reference
Cathode (Graphene family)	LFP@C/G	103.31	500	10 C	89
	GB coated NCMs	205.78 @ 25 °C 195.71 @ 60 °C	100	0.1 C	90
	GO-coated NCMs	136.8	100	1 C	91
	rGO-wrapped NCMs	179.9	100	1 C	92
	LFP/G	142	150	1 C	93

Cathode (MXenes)	d-Ti ₃ C ₂ T _x film based LiFePO ₄	155	70	0.15 C	95
Anode (Graphene family)	Graphene or h-BN on lithium metal anode	~97 % coulombic efficiency	50	2.0 mAh/cm ² and 5.0 mAh/cm ²	99
	LiF/h-BN	>95 % coulombic efficiency	300	0.5 mA/cm ²	100
	3D crumpled graphene balls (CGBs) as scaffolds	97.5 % coulombic efficiency	700	0.5 mA/cm ²	101
	3D current collector based on graphene anchored on copper foam (GN@Cu foam)	154.84	100	0.2 mAh/cm ²	102
Anode (MXenes)	3D porous MXene aerogels as scaffolds (Ti ₃ C ₂)	97.8 % coulombic efficiency	30	3 mAh/cm ²	103
Electrolyte (Graphene family)	LiTFSI/PP13 (ionic liquid)/h-BN	158 @ 150 °C	50	C/8	22
	h-BN/PAN composite nanofibers	133	50	0.1 C	125
	2D boron nitride nanoflakes (BNNF) in gel	82	500	10 C	126

	polymer electrolyte (G-CFBN)				
Electrolyte (MXenes)	2D MXenes containing solid polymer electrolyte (MCSPE)	140 @ 60 °C	50	C/3	127
Electrolyte (Others)	Vertically aligned 2D vermiculite sheets containg solid polymer electrolyte (VAVS- CSPE)	82 % capacity retention @ 35 °C	200	0.5 C	165
Separator (Graphene family)	h-BN coated separator	92 % coulombic efficiency	100	0.5 mA/cm ²	129
	PE-BN/PVDF- HFP bilayer separator	120	500	2 C	130
	GOn coated glass fiber (GF) separator	90 % coulombic efficiency	30	1 mA/cm ²	131
	GO-SiO ₂ modified separator	-	Stable up to 200 cycles Coulombic efficiency up to 350 cycles	1 C	132

Table 3: 2D materials as cell components in Li-S batteries.

Battery component	Material	Reversible capacity (mAh/g)	Cycle number	Current rate	Reference
Cathode (Graphene family)	rGO/S	388	100	0.2 C	144
	Sulfur/activated carbon/graphene (reduced graphene oxide) aerogel (S/AC/GA)	765	100	0.1 C	145
	S-GO/MWCNT paper	997	400	0.2 C	146
	S/PAN/GO	1205	100	0.5 C	147
	BN/graphene/S composites	781.44 @ 70 °C	300	2 C	148
	2D g-C ₃ N ₄ /graphene sheet composite	612.4	1000	1 C	166
	MnO ₂ -graphene nanosheets wrapped mesoporous carbon/sulfur (MGN@MC/S)	600	450	1 C	167
	Porphyrin derived graphene-based nanosheets (PNG)	798	300	0.5 C	168
Cathode (Transition metal dichalcogenides)	MoS ₂ /CNCA/S	107.83	200	0.2 C	150
	MoS _{2-x} /rGO/S	628.2	600	0.5 C	151
	MoS ₂ /S/rGO composite	954	150	0.2 C	152
	MoS ₂ /S nanoflakes	956	300	0.5 C	153
	MoS ₂ @S	632	300	0.5 C	154
	VS ₂ /reduced graphene oxide (G-VS ₂ /S)	532	300	5 C	169
	Co ₄ N@S	1000	100	1 C	170

	Tungsten disulfide and porous carbon nanosheets (WS ₂ /C)	419	500	8 C	171
	S/C-SnS ₂	540	700	2 C	172
	Vanadium nitride nanoribbon/graphene (VN/G)	1252	100	0.2 C	173
	2D MoN-VN/S Heterostructure	555	500	1 C	174
Cathode (MXenes)	MXene/reduced graphene oxide/sulfur	878.4	300	0.5 C	155
	S@TiO ₂ /Ti ₂ C	227.3	200	5 C	156
	3D porous Ti ₃ C ₂ T _x MXene/rGO (MX/G) hybrid aerogel	596	500	1 C	175
	MXene/1T-2H MoS ₂ -C	799.3	300	0.5 C	157
	Crumpled nitrogen-doped MXene nanosheets	610	1000	2 C	176
	2D MXene nanosheets	946.7	200	0.1 C	177
	a-Ti ₃ C ₂ -S/d-Ti ₃ C ₂ /PP	632	50	0.5 C	178
	TiO ₂ QDs@MXene Nanohybrids	680	500	2 C	179
	3D MnO ₂ nanosheets@delaminated-Ti ₃ C ₂ (MNSs@d-Ti ₃ C ₂) aerogel framework	501	500	1 C	180
Cathode (Others)	Exfoliated 2D Lepidocrocite Titanium Oxide Nanosheets	842.34	300	1000	181
Anode (Graphene family)	FGI@Li	94.4	250	1 C	158
	GO-decorated Li anodes	400	800	1 C	159

Anode (Transition metal dichalcogenides)	Li-MoS ₂ anode	940	1200	0.5 C	160
Anode (MXenes)	Ti ₃ C ₂ -Li	750	200	200 mA/g	182
Separators (Graphene family)	ODC/rGO-coated separator	830.0	200	0.1 C	183
	rGO@SL coated separator	523.18	1000	2 C	184
	CeO ₂ /RGO Composite-Modified Separator	859.95	100	0.1 C	185
	ZnO/graphene-functionalized separators	927	200	1 C	186
	BN-carbon separator	780.7	250	0.5 C	187
	BN nanosheet-modified separator	672.3	300	0.2 C	188
	Boron nitride separator	1250	100	0.2 C	189
	GCN-coated-membranes (GCMs)	376.18	200	0.5 C	190
Separators (Transition metal dichalcogenides)	MoS ₂ /Celgard separator	401	600	0.5 C	191
	1T MoS ₂ @CNT	670	500	1 C	192
	MoS ₂ -NP-coated separator	525	150	C/5	193
	MoS ₂ /graphene-coated separator	880	100	0.2 C	194
	MoN _x /Celgard	566	500	0.5 C	195
Separators (MXenes)	Multi-functional separator composed of MXene debris coated eggshell membrane (MXene/ESM)	572	250	0.5 C	196

	Ti ₃ C ₂ T _x -GN interlayer-coated separator	662	200	0.5 C	197
Interlayer (Graphene family)	FBN/G	560	1000	3 C	198
	prGO/S	1013.8	100	0.1 C	199
	PANI-GO	920.53	150	0.5 C	200
	SnO ₂ /rGO	734	200	1 C	201
	PG-Fe ₃ O ₄	356	2000	1 C	202
	EG/Co(OH) ₂	565	300	0.5 C	203
	G@POF-Fe	671	500	0.5 C	204
Interlayer (Transition metal dichalcogenides)	H- and O-incorporated porous TiS ₂ (HOPT)	950	300	0.5 C	205
	MoS ₂ /graphene	720	100	0.2 A/g	206
	rGO/MoS ₂	671	200	0.2 C	207
	MoS ₂ /carbon nanotube	648	500	0.2 C	208
Interlayer (MXenes)	Ti ₃ C ₂ T _x conductive mediator	724	800	0.2 C	209